

Performance of untrenched surface-DAS in near-surface seismic exploration: a comparison with conventional sensors

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SUMMARY

Surface fibre-optic distributed acoustic sensing (S-DAS) typically requires trenching to achieve adequate coupling, which can be challenging and costly in hardrock environments. As an alternative, untrenched S-DAS offers significant time and cost savings, though at the expense of data quality. In this study, we systematically evaluated the impact of untrenched deployment on DAS data quality through a multimethod comparison with conventional sensors, including three-component geophones and vertical-component accelerometers. All recorded data were converted to particle velocity to enable direct amplitude comparisons across arrays. Phase fidelity was assessed using the surface-wave method, cross-power spectral analysis and ambient noise interferometry. Untrenched S-DAS recorded amplitude levels that differed from those of conventional sensors, particularly above 50 Hz. However, within the surface-wave band (up to 35 Hz), the data quality was sufficient to derive reliable shear-wave velocity profiles, especially when the data were resorted to common-receiver gathers. Sensitivity and coherence decreased significantly at higher frequencies and larger offsets, and a systematic time delay was observed for DAS in the surface-wave band. Ambient noise interferometry was ineffective for the untrenched DAS array, largely due to variable channel coupling and system-specific noise. This study provides a systematic field comparison of untrenched S-DAS and conventional sensors in hardrock settings, outlining both the limitations and the practical potential of this cost-effective deployment method.

Key words: Seismic instruments; Seismic interferometry; Seismic noise; Surface waves and free oscillations.

1 INTRODUCTION

1.1 Motivation

The development of fibre-optic sensing has expanded the range of geophysical methods, particularly for measuring vibration, strain and temperature (A.H. Hartog 2017). Distributed Acoustic Sensing (DAS) is increasingly becoming a standard tool in borehole seismics (e.g. T. Daley *et al.* 2015; B. Kuvshinov 2015; R. Pevzner *et al.* 2020) and earthquake seismology (e.g. N.J. Lindsey *et al.* 2017, 2019; I. Lior *et al.* 2021; A. Trabattini *et al.* 2023), owing to its high spatial sampling density and potential for permanent installation. The use of surface-deployed DAS (S-DAS) for near-surface investigations remains an active area of research, with emerging applications in ambient noise tomography and multichannel analysis of surface waves (B. Luo *et al.* 2020; L.

Ehsaninezhad *et al.* 2025). However, its use in reflection seismology and other active-source seismic studies remains unexplored, with only a few documented experiments (K.T. Spikes *et al.* 2019; A. Bakulin *et al.* 2020; B. Brodic *et al.* 2023; M. Al Hasani & G. Drijkoningen 2023; L. Gyger *et al.* 2025; V. Stender *et al.* 2026).

DAS typically refers to a method of measuring dynamic axial strain along an optical fibre using optical interferometry of backscattered light from pulses emitted at an interrogator unit (IU) at one end of the fibre (A.H. Hartog 2017). In other words, DAS is turning an optical fibre (e.g. a telecommunication fibre) into a dense array of longitudinally sensitive single-component relative strain gauges (P. Jousset *et al.* 2018). Common implementations include phase-sensitive Optical Time Domain Reflectometry (Φ -OTDR), chirped-pulse interrogation, phase-coded OTDR and other coherent detection methods (A.H. Hartog 2017).

Seismic exploration methods that rely on amplitude information, such as amplitude variation with offset (AVO), full-waveform inversion (FWI) or horizontal-to-vertical spectral ratios (HVSr), require sensors with a well-characterized and stable amplitude response. Similarly, phase-dependent methods, such as surface-wave dispersion analysis and seismic interferometry, rely on coherent, accurately recorded waveforms. Phase fidelity is also essential for a correct vibroseis sweep correlation. S-DAS, where fibre is usually laid out on the ground and then trenched or covered, enables quick, relatively low-cost deployments. However, this method introduces uncertainty in the mechanical coupling between the fibre and the ground, which can distort the recorded wavefield's amplitude and phase characteristics (N.L. Celli *et al.* 2023; T. Hudson *et al.* 2025). These effects are further amplified by factors such as wave incidence angle, gauge length and site-specific conditions (B. Kuvshinov 2015; I. Lior *et al.* 2021; M.E. Willis 2022).

DAS systems provide several advantages for near-surface seismic applications: they offer a high spatial sampling at the metre scale for Φ -OTDR (A. Yurikov *et al.* 2021; L. Liu *et al.* 2025) or down to the μm scale for other methods (L. Liu *et al.* 2025), enabling an unaliased recording of seismic wavefields. Contrary to conventional moving-mass geophones, the system exhibits a broad-band response (N.J. Lindsey *et al.* 2020; P. Paitz *et al.* 2020), capturing seismic waveforms ranging from long-period scales (up to 200 s; C. Yu *et al.* 2019) to kHz frequencies (L. Liu *et al.* 2025). Additionally, requiring a power supply only at the IU, DAS enables access to locations that are logistically challenging for permanent deployments of conventional sensor arrays, such as boreholes or the seafloor (M.E. Willis 2022). Taking advantage of the axial sensitivity, DAS was successfully used as a borehole sensor for profiling (A. Mateeva *et al.* 2014; T. Daley *et al.* 2015; E. Martuganova *et al.* 2021) and monitoring (R. Pevzner *et al.* 2020), but also as a horizontal surface sensor for beamforming (M.P.A. van den Ende & J.P. Ampuero 2021), Rayleigh and Love wave dispersion measurements (V.R. Tribaldos *et al.* 2021; X. Zeng *et al.* 2021a; Y. Zhao *et al.* 2023), divergence sensor (P. Edme *et al.* 2022), tomographic mine monitoring (X. Zeng *et al.* 2021b) and other ambient noise imaging techniques (B. Benjumea *et al.* 2024; L. Ehsaninezhad *et al.* 2025).

However, the system yields certain limitations for seismic investigations. Transfer of strain from the ground to the fibre depends heavily on the deployment of the fibre, in particular, rock-coupling (B. Kuvshinov 2015; N. Harmon *et al.* 2022; T. Hudson *et al.* 2025) and axial direction relative to the wave incidence (B. Kuvshinov 2015; E.R. Martin *et al.* 2021; M.E. Willis 2022). I. Lior *et al.* (2021) present a case study with DAS signal-to-noise ratio varying along a fibre deployment due to non-uniform site conditions, including temperature. Additionally, the high-wavenumber resolution is limited by gauge-length and pulse-width effects (T. Dean *et al.* 2016b; P. Jousset *et al.* 2018; M.E. Willis 2022; Z. Wilczynski 2025), which impose additional wavenumber filtering on the system's response.

Perhaps the biggest acquisition challenge in using S-DAS for near-surface and engineering studies is the fibre's coupling to the ground. T. Reinsch *et al.* (2017) developed a theoretical framework to analyse the strain transfer from the ground to the fibre, showing that the transfer is frequency-dependent and governed by the lithostatic stress (i.e. the weight of the overburden

material). As a result, shorter seismic wavelengths are less effectively transferred to the fibre, especially in shallow deployments, potentially limiting the high-frequency sensitivity. N.L. Celli *et al.* (2023) provided further insight through full-waveform simulations, demonstrating that the mechanical properties of the material immediately surrounding the fibre (particularly its stiffness) have the most significant impact on DAS recordings. Their results show that the coupling layer can introduce amplitude amplification, phase delays (especially of surface waves), selective amplification of surface-wave phases and generation of interface waves between the medium and the coupling layer. Complementing these theoretical and numerical findings, T. Hudson *et al.* (2025) presented a field case where a fibre initially laid on a glacier surface exhibited poor coupling. However, after melting and refreezing a few centimetres below the surface, the coupling improved significantly, as evidenced by a significant increase in the coherency-based coupling coefficient. These studies highlight that S-DAS coupling is a practical limitation that can vary significantly with deployment conditions.

The most common deployment strategy is to trench the fibre to a depth of several centimetres or more (E. Martin *et al.* 2016; X. Zeng *et al.* 2021a; B. Brodic *et al.* 2023; M. Al Hasani & G. Drijkoningen 2023). However, for loose soils, trenching the fibre does not always ensure a uniform coupling, as shown by V. Stender *et al.* (2026) for mine tailings. Moreover, trenching can be costly and time-consuming, depending on soil conditions, terrain and equipment availability. Particularly in a crystalline rock setting, often characterized by exposed basement or a thin layer of glacial sediments containing boulders of varying sizes, trenching becomes challenging. In contrast to trenching S-DAS, nodal sensors offer fast, reliable deployment without trenching, setting a practical standard for operational efficiency in seismic surveys. Practical alternatives for quick S-DAS deployments are limited. A representative example is demonstrated by B. Brodic *et al.* (2023), who achieved satisfactory coupling by excavating a shallow trench, placing the fibre beneath a thin gravel overburden and consolidating the coupling material by driving a vehicle along the entire cable length. Additionally, N. Harmon *et al.* (2022) conducted a comparative evaluation of rudimentary fibre-coupling techniques, including mechanically pinning the fibre or using a sandbag for coupling. Their results demonstrated that these methods preserved coherent signals relative to co-located geophones, proving the suitability of rapid fibre-optic deployments for near-surface seismic investigations. These findings motivate further investigation into alternative deployment strategies that minimize deployment time and cost while maintaining acceptable data quality.

In this study, we compare the performance of conventional sensors and a surface-deployed DAS laid directly on the ground and covered with gravel, without trenching or mechanical compacting. We began with a theoretical framework for comparing the response of different systems, including the conversion of data to ground velocity. Our analysis was based on field data recorded using a vibroseis source and three different types of sensors: S-DAS, three-component (3C) geophones and vertical-component microelectromechanical system (MEMS) accelerometers. Our analysis included: (1) a quantitative comparison of root-mean-square (RMS) ground velocity amplitudes across recording systems and frequency bands; (2) a quantitative comparison of amplitude and phase information through cross-power spectrum and its associated properties; (3) an evaluation

of surface-wave phase information through phase-shift multi-channel analysis of surface waves (MASW; C.B. Park *et al.* 1999) and (4) a comparison of cross-correlation functions of ambient noise. By addressing both amplitude and phase fidelity, we provide complementary insights for exploration practitioners considering untrenched S-DAS deployment.

1.2 Nature of the recording arrays

1.2.1 DAS

In this work, we focus on a Φ -OTDR DAS system. It emits a single coherent, monochromatic light pulse into the fibre and analyses the Rayleigh backscattered signal over a fixed gauge length to detect dynamic strain. Backscattering arises from small variations in the fibre's refractive index, caused by intrinsic heterogeneities in the glass. The phase of the returned signal depends on the optical path length, which is the effective distance light travels within the fibre. This makes the phase measurement sensitive to strain (or deformation) in the interrogated fibre. We revisit the theory of DAS under the assumption of a perfectly rectilinear fibre with ideal ground coupling. The limitations of this assumption are addressed later in the paper.

Following the nomenclature of A. Trabattini *et al.* (2023), the deformation $\delta(s, t)$ at curvilinear position s along the fibre and at time t is defined as the difference between the deformed fibre length $L(s, t)$ (e.g. due to an incident seismic wavefield) and the reference length $L_0(s)$. This deformation corresponds to the cumulative strain ε along the fibre, expressed as the spatial integral:

$$\delta(s, t) = L(s, t) - L_0(s) = \int_0^s \varepsilon(s', t) ds'. \quad (1)$$

Rayleigh backscatter from a position s in the fibre when at rest is characterized by a reference phase $\phi_0(s)$, representing the static optical path. When the fibre deforms due to external perturbations, the optical path length relative to the IU changes, inducing a time-dependent phase shift. The total recorded phase becomes:

$$\phi(s, t) = \phi_0(s) + \frac{4\pi n\xi}{\lambda} \delta(s, t) = \phi_0(s) + K\delta(s, t). \quad (2)$$

Here $K = \frac{4\pi n\xi}{\lambda}$ is a system-specific scaling factor that converts the change in optical path $\delta(s, t)$ to phase, with λ the central wavelength, n the effective refractive index and ξ the photoelastic coefficient for longitudinal strain in isotropic material (A.H. Hartog 2017). In practice, DAS systems record the time derivative of the phase difference, referred to as the phase rate, $\dot{\phi}(s, t)$. Taking the time derivative of the measured phase $\phi(s, t)$ eliminates the static reference phase $\phi_0(s)$ and allows efficient data storage (e.g. in 16-bit format ranging from $-\pi$ to π). The phase rate $\dot{\phi}$ is directly proportional to the deformation rate $\dot{\delta}(s, t)$. Additionally, as shown in eq. (1), deformation is a spatial integral of the strain component along the fibre. Accordingly, the deformation rate $\dot{\delta}$ is a spatial integral of the strain rate $\dot{\varepsilon}$.

$$\dot{\phi}(s, t) = K \cdot \dot{\delta}(s, t) = K \int_0^s \dot{\varepsilon}(s', t) ds'. \quad (3)$$

The spatial derivative of $\dot{\phi}(s, t)$ is referred to as a differential phase rate (A.H. Hartog 2017), and allows direct recovery of the strain rate $\dot{\varepsilon}$. In practice, the spatial derivative of phase (differential phase) is estimated using a central finite difference over a

spatial averaging window called the gauge length (GL), giving:

$$\frac{1}{K} \frac{\partial \dot{\phi}}{\partial s}(s, t) \approx \frac{\dot{\delta}(s + \frac{GL}{2}, t) - \dot{\delta}(s - \frac{GL}{2}, t)}{GL} \approx \dot{\varepsilon}(s, t). \quad (4)$$

As presented in eq. (3), the Φ -OTDR-based DAS natively measures the phase $\phi(s, t)$ or its temporal derivative—phase rate $\dot{\phi}(s, t)$. Further, using the gauge length central finite difference approximation from eq. (4), the measurement of deformation rate $\dot{\delta}(s, t)$ is evaluated over the GL window, and a strain rate $\dot{\varepsilon}(s, t)$ is recovered. This approach allows retrieval of local information on the recorded strain, effectively transforming the fibre from a transmission-based system into an array of independent strain-rate sensors, each resolved over a single gauge length.

The operation presented in eq. (4) acts as a spatial moving average, which is equal to a convolution with a boxcar function (with a scaling factor). The boxcar function in the wavenumber k domain is represented by $\frac{\sin(k)}{k}$, commonly described as a sinc function (T. Dean *et al.* 2016a; M.E. Willis 2022). Thus, after taking the Fourier transform, convolution with the boxcar becomes multiplication by the sinc function. This results in a spatial transfer function in the wavenumber domain k (where $k = \frac{2\pi f}{c}$), and the strain rate measured by DAS becomes:

$$\dot{\varepsilon}^{\text{DAS}} = \dot{\varepsilon} \text{sinc}\left(k \frac{GL}{2}\right) = \dot{\varepsilon} \text{sinc}\left(\frac{\pi f}{c} GL\right). \quad (5)$$

Increasing the gauge length provides a signal with a higher signal-to-noise ratio (SNR) by effectively averaging over a longer fibre segment (T. Dean *et al.* 2016a; M.E. Willis 2022), which reduces high-frequency noise. However, this spatial averaging comes at the cost of reduced wavenumber resolution (N.J. Lindsey *et al.* 2020; M.E. Willis 2022). As a result, the waveform amplitude and phase are affected depending on the frequency f and wave velocity c . Specifically, high-frequency, low-velocity signals experience greater attenuation as the gauge length GL increases, with amplitude zero crossings at specific wavenumbers that depend on GL.

Due to this SNR and resolution trade-off, the optimal gauge length depends on the application, source and medium parameters. In general, it can be assumed that a gauge length of 10 m provides a flat response for a typical seismic wavefield range of frequencies and velocities (P. Jousset *et al.* 2018). In Fig. 1(a), we present the transfer function of a system using a 10 m long gauge length for a wave velocity c equal to 2500 m s⁻¹, calculated using eq. (5). The derived amplitude attenuation is not significant and does not present sinc zero-crossings (notches) in the frequency band below 100 Hz. Evaluating lower velocities or introducing a longer gauge length would narrow the main lobe of the sinc function and introduce zero crossings at lower frequencies, degrading response fidelity for high-frequency signals. The phase curve is shifted by $-\pi/2$ to represent the phase difference relative to a velocity measurement, which arises from spatial integration of velocity. We refer to T. Daley *et al.* (2015), M.E. Willis (2022) and J. Yang *et al.* (2022) for a detailed analysis of the gauge length effect.

Another intrinsic factor that shapes the DAS amplitude response and spatial resolution is the width of the optical pulse. While it is common to model DAS data as the difference between two discrete phase measurements, in practice, the measurement process effectively applies two Gaussian windows, corresponding to two optical pulses with approximately Gaussian envelopes (T. Dean *et al.* 2016b). Nevertheless, as noted by T. Dean *et al.*

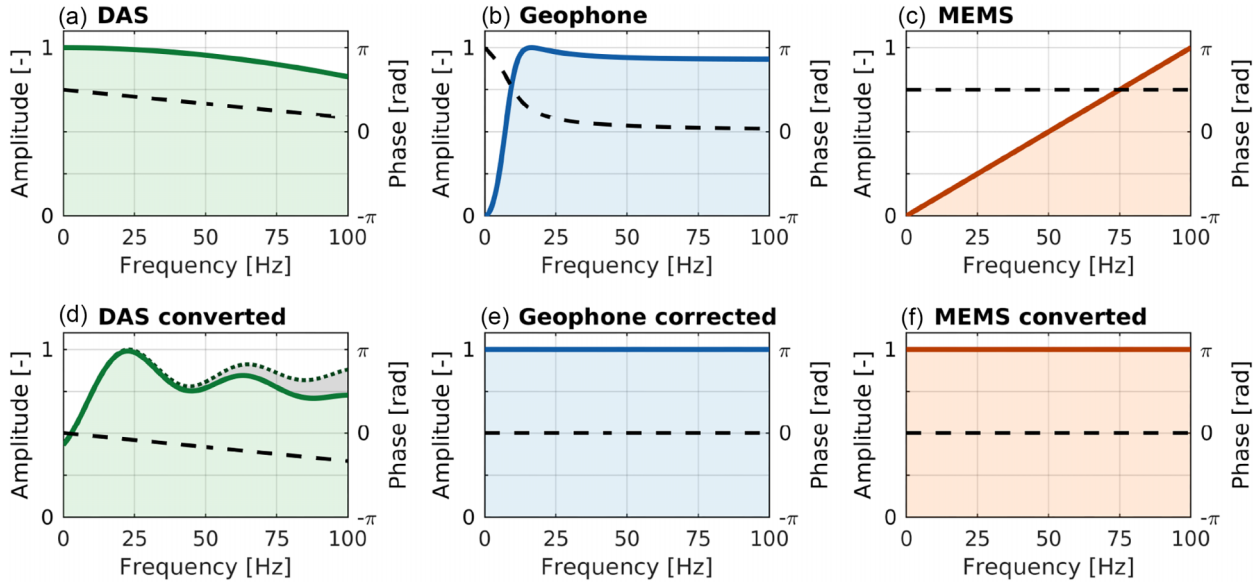


Figure 1. Instrument amplitude and phase response of the sensors in their native domain (top row) and after conversion to velocity (bottom row). Amplitudes are normalized. Dashed lines represent phase response (right axis). Presented for DAS (a, d), geophones (b, e) and accelerometers (c, f). The DAS transfer function is plotted for an example wave velocity, c , of 2500 m s^{-1} . The dotted line in plot (d) corresponds to the amplitude scaling due to the Hann window average removal.

(2016a), this effect becomes negligible provided the pulse width is less than half of the gauge length.

1.2.2 Geophones

A standard moving-mass geophone is a sensor consisting of a magnetic mass suspended by a spring and moving within a fixed reference coil. The output voltage is proportional to the relative velocity between the mass and the coil. Importantly, the motion of the mass is damped to avoid excessive oscillations in the resonance frequency, which depends on the stiffness of the spring and the weight of the mass. The system is characterized by three key parameters: the natural frequency of the free-swinging system, f_0 ; the damping constant, h ; and the sensitivity, S , expressed in $\text{V (m s}^{-1})^{-1}$, which reflects the sensitivity of the recorded voltage to ground velocity (J. Havskov & L. Ottemöller 2010, eq. 3.9). These parameters influence the instrument response. An exemplary 10 Hz geophone transfer function is shown in Fig. 1(b), illustrating the system's effects on the amplitude and phase of the recorded signal.

1.2.3 MEMS accelerometers

The accelerometers used in this study are MEMS-based. Similar to geophones and other accelerometers (J. Havskov & L. Ottemöller 2010), MEMS-based devices detect the motion of a proof mass in response to vibration. The proof mass, micro-machined as a thin silicone slice coated with metal, acts as a movable electrode. As the sensor experiences acceleration, the change of capacitance is recorded as a voltage signal by the sensor (J. Laine & D. Mougenot 2014).

A key advantage of MEMS accelerometers is their flat amplitude and phase response across wide frequency ranges. Unlike geophones, which introduce phase shifts due to their damped mechanical design, MEMS accelerometers exhibit no phase distortion. They can detect very low-frequency signals, including

static (DC) accelerations such as tilt or gravity (N. Tellier *et al.* 2020). However, in exploration seismology, a high-pass filter is typically applied to suppress low-frequency noise and remove DC components. Accelerometers are generally considered superior to geophones due to their flat response, broad operational bandwidth, low self-noise levels and sensitivity to low frequencies (J. Laine & D. Mougenot 2014; N. Tellier *et al.* 2021).

Fig. 1(c) shows the normalized amplitude and phase response of a MEMS accelerometer, both presented relative to those of a velocity sensor. The amplitude is scaled by $\frac{1}{\omega}$, highlighting the accelerometer's increased sensitivity to high-frequency signals compared to velocity measurements. The phase response includes a $\pi/2$ offset, representing the phase shift introduced by integrating acceleration to obtain velocity.

1.2.4 Comparability of the systems

The three seismic sensors used in this study—the geophones, the accelerometers and DAS—record the same seismic wavefield, but the output corresponds to different ground motion kinematic parameters: velocity $v(t)$, acceleration $a(t)$ and strain rate $\dot{\epsilon}(t)$. To compare the systems' responses, the recorded signals need to be converted to a uniform unit.

The conversion of the ground motion time-series—being the ground displacement $u(t)$, ground velocity $v(t)$ or ground acceleration $a(t)$ —is seemingly trivial mathematically, as they are just temporal transformations of each other. The conversion from acceleration $a(t)$ to velocity $v(t)$ is simply integration in time and is expressed as follows in the frequency domain:

$$V(\omega) = \frac{1}{i\omega} A(\omega). \quad (6)$$

Here, $V(\omega)$ and $A(\omega)$ are the Fourier transformations of $v(t)$ and $a(t)$, respectively. However, as $\omega \rightarrow 0$, this transformation will amplify low frequencies and approach infinity, requiring high-pass filtering or regularization. Additionally, any bias in the

acceleration data will accumulate over time, leading to a diverging output. Thus, it no longer presents a trivial problem in real seismic data processing, and has been a matter of scientific discussion (S. Thenozhi *et al.* 2012, and references therein).

Comparison of strain (or strain rate) data with the rest of the ground motion time-series requires a transformation in space. Strain ε is the spatial derivative of the displacement vector $\mathbf{u}$, which makes the strain rate $\dot{\varepsilon}$ a spatial derivative of the velocity vector $\mathbf{v}$. Therefore, the sensor comparison can be performed either in velocity, after spatially integrating the strain rate data, or in strain rate, after spatially differentiating the velocity data.

Differentiating the velocity data in space would simulate the virtual strain rate gauges between consecutive sensors (S. Zulic *et al.* 2022), and thus requires a sensor spacing equal to the gauge length used in DAS measurements. This is, in practice, difficult to achieve due to the uneven spacing of the sensors in the field and would result in strain rate estimates that are not comparable to the DAS recordings.

A common way to convert strain data to a geophone equivalent is based on the derivations by T. Daley *et al.* (2015), who proved that for a single plane wave propagating along a rectilinear fibre with an apparent velocity c , the velocity measurement is obtained from strain through the relationship:

$$c\varepsilon = \pm v, \quad (7)$$

which is also applicable for strain-rate $\dot{\varepsilon}$ and acceleration a . Further method expansions incorporate frequency-wavenumber ($f - k$) rescaling to simultaneously integrate in space and differentiate in time (T. Daley *et al.* 2015; H.F. Wang *et al.* 2018; N.J. Lindsey *et al.* 2020). When rescaling wavenumbers near zero, the method requires introducing a water level for stability (J. Correa *et al.* 2017), which may introduce unphysical modifications to the phase and velocity (P. Paitz *et al.* 2020).

The rescaling method is based on an assumption of a rectilinear fibre, for which a deformation $\delta(s, t)$ is equal to a displacement at a point s relative to the displacement at the beginning of the cable (A. Trabatttoni *et al.* 2023):

$$\delta(s, t) = u(s, t) - u(0, t). \quad (8)$$

If the displacement at the reference point $s = 0$ is zero, then the deformation $\delta(s, t)$ is equal to the displacement $u(s, t)$. However, if the reference displacement $u(0, t)$ is non-zero, it introduces a spatially constant offset into the measurements. This offset, often referred to as interrogator noise, arises from vibrations at the interrogator unit during acquisition and is commonly observed in seismic data (S. Dou *et al.* 2017; H.F. Wang *et al.* 2018; N.J. Lindsey *et al.* 2020; M.E. Willis *et al.* 2021). Also known as common-mode noise, this disturbance manifests in strain or strain-rate measurements and propagates into velocity data as a cumulative bias.

However, in the case of a segment-wise rectilinear or curvilinear geometry, the accumulation of the constant bias (offset) becomes more complex. As mathematically derived by A. Fichtner *et al.* (2022) and further evaluated and illustrated by A. Trabatttoni *et al.* (2023, figs 1 and 2), the deformation of a curved fibre depends on the sharpness in the change of the fibre direction, introducing non-local effects represented as horizontal, space-invariant features.

A. Trabatttoni *et al.* (2023) proposed a conversion method suitable for both segment-wise rectilinear and curvilinear geometry. They converted the strain data to deformation via spatial integration and applied a weighted sliding-window spatial average

to remove non-local effects arising from changes in fibre curvature. The authors (A. Trabatttoni *et al.* 2023) recommend using a cumulative sum integration, followed by a sliding Hann window for weighted-average removal due to its favourable spectral characteristics, which reduce low-wavenumber resolution and further affect the amplitude response.

Thus, when converting the strain-rate data to velocity, the final DAS instrument response depends on the gauge-length effect (sinc filtering), spatial integration and weighted moving-average filtering. This combined effect is illustrated for an example velocity of $c = 2500 \text{ m s}^{-1}$ in Fig. 1(d), for a gauge length GL of 10 m and a Hann window of 150 m. The phase is shifted by $-\pi/2$ due to spatial integration. The Hann window filter, marked with a dotted line, introduces an average amplitude scaling of approximately 0.85 across frequencies. Importantly, this average value is consistent across all velocities, although the positions of the side lobes vary with wavenumber and window size. Using a longer averaging window would shift the first lobe closer to zero frequency, allowing the recovery of lower wavenumbers. Because of the filter's symmetry, the signal's phase is not modified. We refer to A. Trabatttoni *et al.* (2023) for a detailed analysis of the weighting function and window length effects.

Finally, to enable direct comparison of the retrieved DAS velocity with the other sensors, the geophone data has to be corrected by a complex instrument response, which results in a flat phase and amplitude response (Fig. 1e). The accelerometer data needs to be integrated to velocity, which introduces a $-\pi/2$ phase shift (Fig. 1f), and scales the amplitudes by $\frac{1}{\omega}$. The low-frequency amplification, resulting from division by frequencies approaching zero, is corrected by modern integrators (e.g. S. Thenozhi *et al.* 2012) which use baseline corrections and high-pass filtering. The schematic response shown in Fig. 1(f) illustrates that, after conversion, the accelerometer signal becomes directly comparable to velocity measurements in a certain frequency band.

We conclude this subsection with a visual synthesis of the comparability framework. Fig. 1 shows the instruments' transfer functions in their native recording domains (a–c) and following conversion to particle velocity (d–f), enabling direct evaluation of instrument responses. Fig. 2 provides a comprehensive unit conversion chart. For simplicity, we considered a rectilinear DAS deployment, assuming that the deformation equals the displacement, as in eq. (8).

2 EXPERIMENTAL SITE AND DATA

The experiment was conducted on the premises of the inactive Blötberget iron oxide mine. The mine is located in the Bergslagen ore province in central Sweden, a major mining district in Sweden and Europe. The Blötberget mine has served as a geophysical test site for hardrock mineral exploration for more than a decade (see A. Malehmir *et al.* 2020, and references therein). Throughout the years, seismic campaigns have been conducted using a variety of sensors, sources and methods. The experiment presented in this paper is part of a seismic campaign conducted in June 2022. The instrumentation included collocated arrays of accelerometers, three-component (3C) geophones and DAS, including both untrenched S-DAS and a borehole horizontally wound cable (HWC) (L. Gyger *et al.* 2024). The instrument geometry was set up for reflection seismic imaging of the main mineralization using broad-band sensors and surface DAS (L.

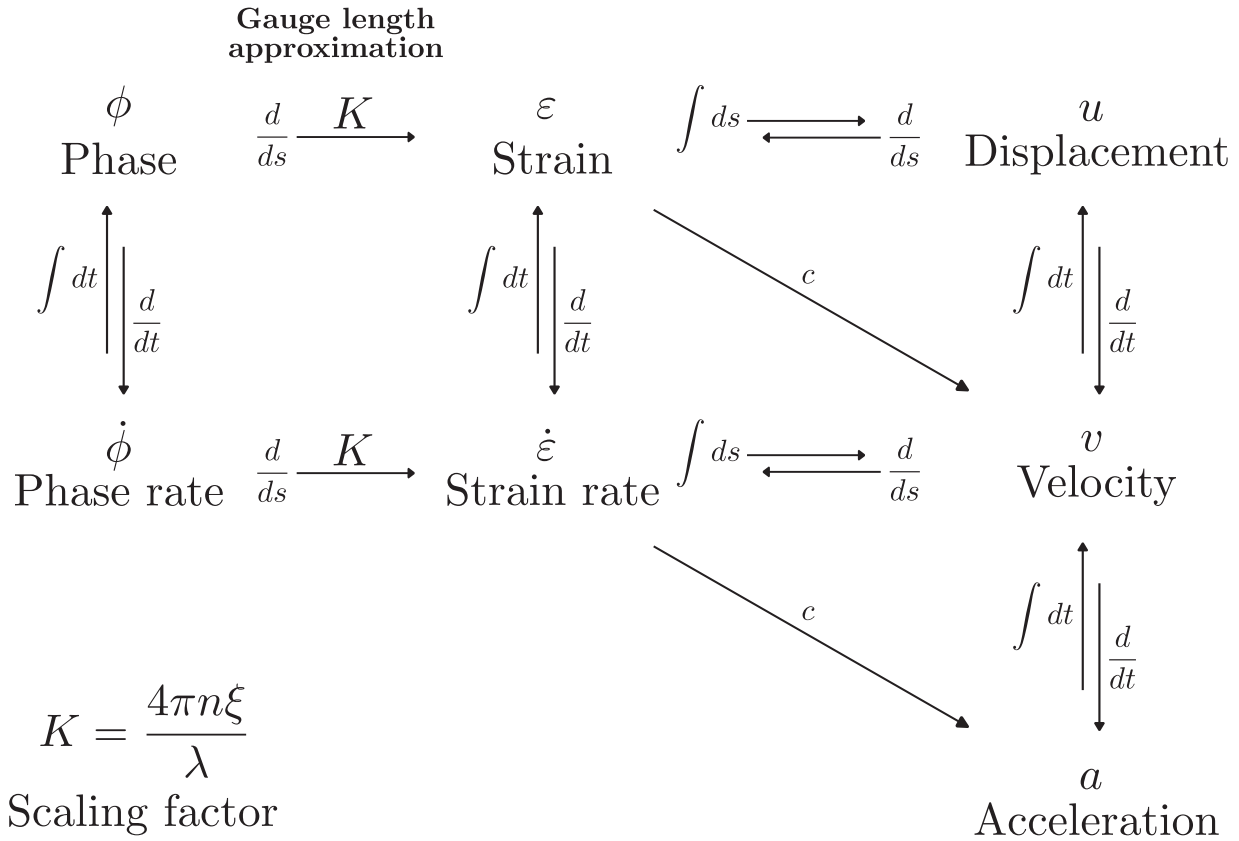


Figure 2. Unit conversion chart with the assumption of a perfectly coupled rectilinear fibre, simplifying $\partial/\partial s$ to d/ds .

Gyger *et al.* 2024, Gyger, 2025). Complementary studies focused on passive seismic surface-wave imaging (Z. Wilczynski *et al.* 2024), comparison of the sensors' performance (Z. Wilczynski *et al.* 2025), multicomponent imaging (M. Markovic *et al.* 2025) and ambient noise characterization (A. Kaslilar *et al.* 2022).

The geology of the area is characterized as hardrock, with a shallow crystalline basement and a thin layer of surface glacial sediments, reaching 20 m in thickness. The near-surface bedrock reveals a velocity gradient, most likely due to compressional fracturing or weathering (Z. Wilczynski *et al.* 2024). Simplified average seismic velocities of the site are presented in Table 1.

In this study, we focus on active-source measurements using collocated arrays of S-DAS, nodal three-component geophones and nodal MEMS-based accelerometers. The arrays had different lengths due to the number of available sensors. The acquisition was conducted in 2-D geometry with a nearly linear profile extending from NW to SE (Fig. 3). The sensors were primarily deployed along a forest dirt road. The array parameters are presented in Table 2. The north component of the geophones was pointing along the seismic profile, which is why we refer to it as a radial component (3CH1).

The seismic source used in the experiment was a vibrotruck using a linear sweep from 2 to 200 Hz. Source parameters are given in Table 2. In the northern part of the profile, due to bad road conditions, the base plate could not be coupled to the ground, and around 70 shot points had to be skipped. Therefore, the total number of recorded shot points was 420.

During deployment and acquisition, we experienced several challenges. First of all, due to outcropping bedrock and coarse gravel in some locations, the coupling of the nodal sensors was problematic. MEMS sensors equipped with spikes were fairly easy to plant, but 3C geophones, built in a large, rounded casing, had to be dug into the ground, which could have affected coupling quality. Additionally, some nodal units connected to the 3C geophones experienced performance issues, with parts of the array not recording during the acquisition.

The DAS fibre was deployed by the field crew. It was unrolled from the drum, laid out along the road and then covered with a thin layer of gravel using an excavator (Fig. 3). The deployment took less than a day but did not achieve optimal fibre coupling, with surface conditions varying between locations. The interrogator was installed in a shed on a private semi-industrial property, and some activities during acquisition introduced noise to the measurements. A tap test was performed at several points along the profile to assign coordinates to the fibre length. Sparse test points (several hundred metres apart) provide a good approximation for a nearly linear fibre deployment, but may introduce inaccuracies in the sensor locations.

The abovementioned deployment and acquisition challenges affected the data quality. After visually inspecting the data, we narrowed our investigation to collocated sensors that remained active throughout acquisition and recorded data of satisfactory quality, with distinguishable wavefield components. We selected eight stations (Fig. 3, black scatter points) and resorted the data into common-receiver gathers. Each gather contained 420 traces, resulting in 3360 shot-station pairs. For each shot point,

Table 1. A simplified 1-D velocity model of seismic velocities in Blötterget site. Depth represents the depth to the top of each layer. The velocities in the weathered bedrock are assumed to follow a linear increase. Modified from Z. Wilczynski *et al.* (2024).

Layer	Depth (m)	V_p (m s ⁻¹)	V_s (m s ⁻¹)	ρ (kg m ⁻³)
Glacial sediments	0	1200	500	2200
Weathered bedrock	0–20	5170	2600	2650
Increase per metre depth		+2.77	+1.80	+0.3
Bedrock	300	6000	3140	2750



Figure 3. Survey location map (left) with an inset on the points selected for investigation; Field photo with the acquisition layout (centre). The thin layer of gravel along the road covers the fibre; a schematic sketch of the acquisition (right).

Table 2. Seismic survey parameters. For DAS, temporal sampling corresponds to the pulse repetition frequency.

Data	DAS	3C	MEMS
Sensor type	Φ -OTDR	3-component geophone (10 Hz)	MEMS-based accelerometer
Model	AP Sensing DAS N51	Sercel SG-10HS	Sercel WiNG
Profile length (m)	2190	1490	2455
Sensor spacing (m)	5	10	5
Gauge length (m)	10	–	–
Temporal sampling (Hz)	20 000	500	1000
Seismic source			
Source type	Vibroseis		
Model	Nomad 15, 77 kN peak force		
Source sweep	Linear, 2–200 Hz		
Sweep duration	18 s		
Sweep + listening time	18 + 7 s		
Shot point spacing	5 m, at every MEMS station		
Sweeps per point	min. 3		

we used three repeated shots, yielding more than 10 000 traces in total.

Throughout this paper, we use the naming convention presented in Table 2: DAS for surface DAS, MEMS for MEMS-based accelerometers and 3C for a three-component geophone; specifically 3CV for vertical component, 3CH1 for radial component and 3CH2 for transverse component (normal to the acquisition profile).

The raw data (native unit) common receiver gathers, cross-correlated with the theoretical source-sweep, are presented in Figs 4(a)–(c) for 3CH1 (a) and DAS (b).

3 METHOD

We compared the data recorded by the different systems in several aspects. First, we analysed amplitudes converted to a uniform ground motion parameter: ground velocity. Next, we quantified the similarity between sensors using magnitude-squared (MS) coherence and analysed their phase relationships by calculating phase differences and time delays from the cross-power spectra. Then, as a proxy for phase coherence across data sets, we applied surface-wave analysis methods, which rely on a phase agreement of signals. Finally, for the DAS and MEMS data sets, we attempted to retrieve surface waves from noise correlations.

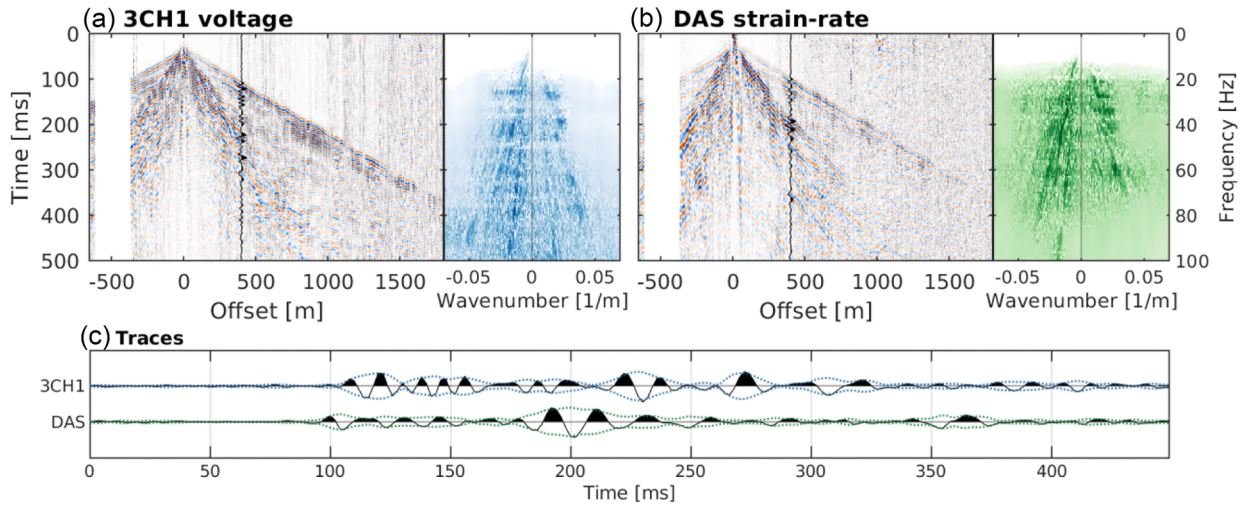


Figure 4. Common receiver gathers in native receiver units, cross-correlated with a theoretical sweep (2–200 Hz, 18 s duration) and normalized, recorded by: (a) 3CH1 in voltage; (b) DAS in strain rate. For each panel, the plot on the right shows the corresponding f - k spectrum of the gather. All colour scales are normalized. The black wiggle traces overlaid on the gathers correspond to the individual traces shown in panel (c). Dotted lines show the signal envelopes.

With amplitude information removed during pre-processing, we used it as an additional proxy for phase agreement. Throughout the analysis, we used uncorrelated vibroseis data to avoid mixing the kinematic ground motion parameters of the theoretical sweep with sensor responses.

3.1 Ground velocity analysis

To retrieve velocity measurements, we first converted each data set from its native units, as detailed below. For all traces, we applied a 10 per cent Tukey window and a second-order zero-phase Butterworth bandpass filter, with corner frequencies tailored to each analysis. The processing and conversion workflow is summarized in Table 3 and described in the following subsections.

3.1.1 MEMS-based accelerometers

Acceleration data were recorded in millivolts. After decimating to 500 Hz, we converted the signals to volts, detrended them and applied a 10 per cent Tukey window. We then divided the data by a constant sensitivity factor of $0.452 \text{ V m}^{-1} \text{ s}^2$ (N. Tellier *et al.* 2021) to recover acceleration. Data were subsequently filtered to the desired frequency band.

To convert acceleration to velocity, we used the algorithm proposed by S. Thenozhi *et al.* (2012), implemented in the MATLAB `convertVibration` package. The method includes offset calibration, low-pass filtering (acting as a numerical integrator), baseline correction and high-pass filtering. The resulting velocity signal is characterized by a $-\pi/2$ phase shift and amplifies lower frequencies with respect to the original acceleration signal (Figs 1c and f).

3.1.2 Geophones

Geophones are sensitive to ground velocity and output signals in millivolts. After converting to volts, we applied detrending, windowing and a bandpass filter to isolate the relevant frequency

range. Following eq. (3.9) of J. Havskov & L. Ottemöller (2010), we corrected for the instrument response in the frequency domain by dividing the Fourier-transformed data by the instrument response function obtained using the geophone parameters: natural frequency $f_0 = 10 \text{ Hz}$, damping ratio $h = 0.56$ and sensitivity $S = 85.8 \text{ V (m s}^{-1})^{-1}$. A visual representation is shown in Figs 1(b) and (e). To mitigate drift caused by low-frequency amplification (similar to numerical integration artefacts), we applied additional filtering and detrending.

3.1.3 DAS

Raw DAS data were recorded in differential phase rate. As detailed earlier, we converted the data to strain rate using a gauge length of 10 m. In the analysed frequency band (up to 100 Hz) and for wave velocities above 2000 m s^{-1} , the gauge length wavenumber scaling introduced no zero crossings and only minor amplitude decay at higher frequencies (Fig. 1a). The resulting strain rate represents the temporal change of the strain along the fibre.

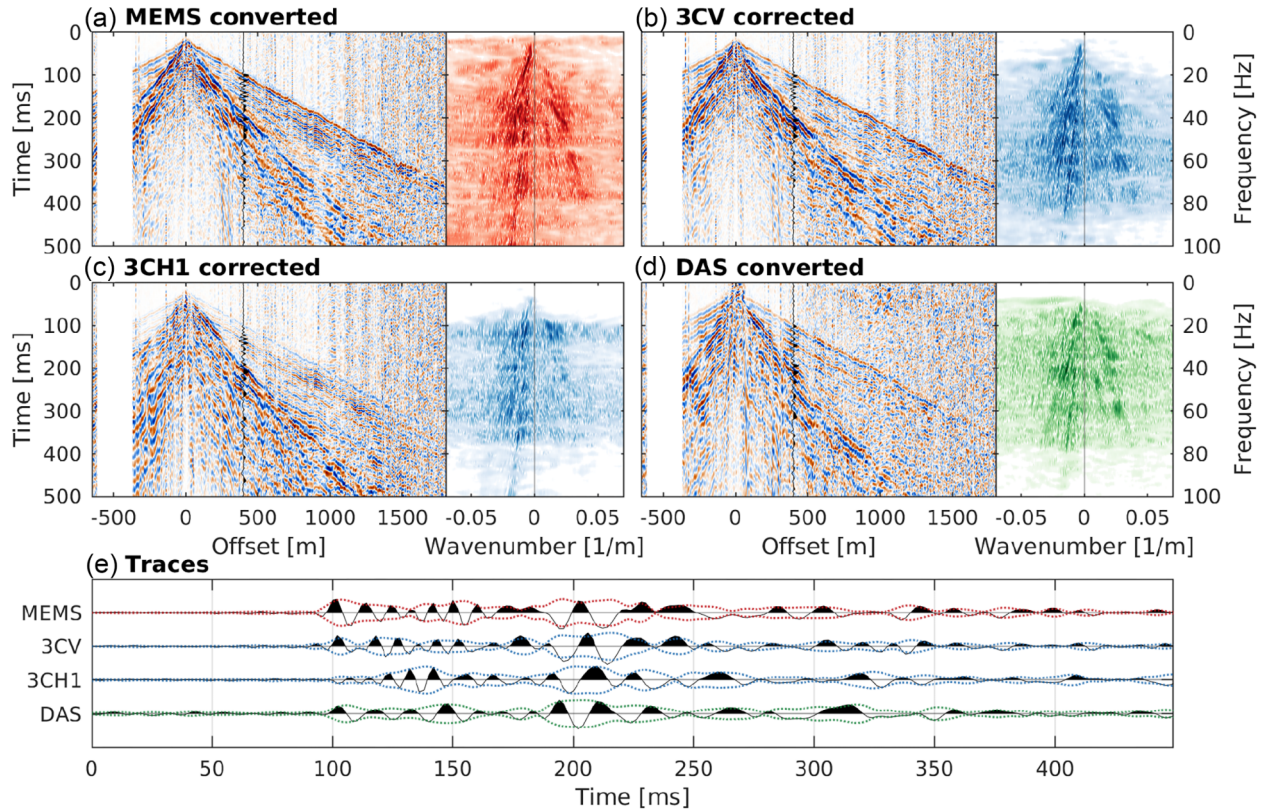
After windowing, decimating and detrending the signals, we applied the spatial integration and weighted moving average method of A. Trabattini *et al.* (2023) presented in Section 1.2.4 to obtain ground velocity. We used a 150 m Hann weighting window as a sliding mean filter, which introduced additional wavenumber filtering (Fig. 1d). The symmetric Hann filter preserved phase relationships but reduced amplitudes by an average factor of 0.85, with the exact scaling depending on the particular wavenumber and, thus, on the signal frequency and apparent velocity. Finally, the resulting particle velocity data were filtered to the frequency bands of interest.

3.1.4 Comparability

We visually compared the converted data sets to ensure the process was stable. In Figs 5(a)–(d), we plotted the common receiver gathers with their f - k spectra after correcting for the instrument response (3CV, 3CH1) and converting to velocity (MEMS,

Table 3. Processing sequences to convert the data sets from their native recording quantity to ground velocity.

Step	MEMS	Unit	3C	Unit	DAS	Unit
1	Decimate to 500 Hz	mV	–	mV	Decimate to 500 Hz	rad s ⁻¹
2	Convert to volts	V	Convert to volts	V	Convert to strain rate	s ⁻¹
3	Detrend	V	Detrend	V	Detrend	s ⁻¹
4	Tukey window (10 per cent)	V	Tukey window (10 per cent)	V	Tukey window (10 per cent)	s ⁻¹
5	Convert to acceleration	m s ⁻²	Butterworth filter	V	Convert to deformation rate	m s ⁻¹
6	<i>Sensitivity correction</i>		–		<i>Integrate in space</i>	
7	Convert to velocity	m s ⁻¹	Instrument response correction	m s ⁻¹	Convert to velocity	m s ⁻¹
8	<i>Integrate in time</i>				<i>Moving mean filter</i>	
7	Butterworth filter	m s ⁻¹	Butterworth filter	m s ⁻¹	Butterworth filter	m s ⁻¹
8	–		Detrend	m s ⁻¹	–	

**Figure 5.** Common receiver gathers after correcting for the instrument response (geophones) and converting to velocity (MEMS, DAS), cross-correlated with a theoretical sweep (2–200 Hz, 18 s duration) and normalised, recorded by: (a) MEMS. (b) 3CV. (c) 3CH1. (d) DAS. For each panel, the plot on the right shows the corresponding f–k spectrum of the gather. All colour scales are normalised. The black wiggle traces overlaid on the gathers correspond to the individual traces shown in panel (e). Dotted lines show the signal envelopes.

DAS). The particle data were cross-correlated with the theoretical source sweep (2–200 Hz, 18 s duration; Table 2). In Fig. 5(e) we compared single traces of the same source–receiver geometry.

Retrieving a uniform ground-motion parameter (i.e. velocity) enabled direct, trace-by-trace comparison of signal amplitudes. For each collocated pair of receivers, we identified source–receiver pairs and calculated RMS amplitudes within specific frequency bands. We analysed several sweep intervals and applied bandpass filters accordingly: low-frequency sweep (0 to 4 s, 5 to 35 Hz); high-frequency sweep (2 to 12 s, 55 to 100 Hz) and post-sweep noise (20 to 24 s, both frequency bands).

This approach enabled frequency-dependent analysis of waveform amplitudes: the 5–35 Hz band captures mostly surface-wave energy, and the 55–100 Hz band is more sensitive to body waves. Noise was analysed using the listening window following the sweep (20–24 s). Frequencies below 5 Hz were excluded to ensure stable velocity conversion.

By calculating the root-mean-square (RMS) amplitude for each source–station pair, we analysed the absolute energy recorded by the sensors in each trace. After visually inspecting the cross-correlated data (Figs 5a–e), we restricted the analysis to offsets between 100 and 1000 m, matching the offset range used in the subsequent surface-wave analysis. This yielded more than 5000 valid traces. We analysed the data by computing the ratio

of RMS amplitudes between recording systems and derived basic statistical measures. The mode was estimated by binning the data into 0.1-wide linear intervals and selecting the bin with the highest count.

3.2 Coupling coefficient

Following T. Hudson *et al.* (2025, eq. 5), we quantify fibre-medium coupling using a channel-to-channel coherency metric based on the normalized zero-lag correlation. For a time window of length t_{win} , the coupling coefficient between neighbouring traces i and $i + 1$ is defined for strain ε as:

$$C_i = \sum_{t=0}^{t=t_{\text{win}}} \frac{\varepsilon_i(t) \cdot \varepsilon_{i+1}(t)}{|\varepsilon_i(t)| \cdot |\varepsilon_{i+1}(t)|}. \quad (9)$$

The same formulation can be applied to other observables (e.g. strain rate $\dot{\varepsilon}$, particle velocity v or acceleration a). The final coupling estimate is obtained by computing C_i for successive time windows and averaging over all selected windows.

We computed coupling coefficients using one hour of night-time ambient-noise recordings from the DAS and MEMS arrays, with non-overlapping 1 s windows. To emphasize spatially persistent coupling variations, we also smoothed the channel-wise coefficients using an 11-point moving average (corresponding to a 50 m window). Values near 0 indicate no similarity, whereas values close to 1 indicate strong similarity between neighbouring traces, and thus, comparatively good coupling. We did not use the geophone array due to inconsistent channel spacing and sensor performance issues.

3.3 Coherence and phase analysis

A complex cross-power spectrum $P_{xy}(\omega)$ is the Fourier transform of two cross-correlated time-series $x(t)$ and $y(t)$ (A.G. Piersol & J.S. Bendat 1993). We used it to quantify the phase similarity between sensors after converting the signals to a common unit.

The argument of the complex cross-power spectral density $P_{xy}(\omega)$ gives the phase difference $\phi_{xy}(\omega)$ between the two signals at each frequency:

$$\phi_{xy}(\omega) = \arg\{P_{xy}(\omega)\}. \quad (10)$$

After phase unwrapping, the phase difference can be converted to a time delay by dividing by the radial frequency ω , providing an estimate of the relative time delay between the signals at a given frequency:

$$\tau_{xy}(\omega) = \frac{\phi_{xy}(\omega)}{\omega}. \quad (11)$$

Magnitude-squared (MS) coherence C_{xy} is a quantity related to the cross-power spectrum and is a measure of the linear relationship between two signals (G.C. Carter 2005). It is defined as the magnitude of $P_{xy}(\omega)$, normalized by the power spectral densities of the two input signals:

$$C_{xy}(\omega) = \frac{|P_{xy}(\omega)|^2}{P_{xx}(\omega)P_{yy}(\omega)}. \quad (12)$$

MS coherence quantifies the correlation between the power of two signals, with values ranging from 0 (no correlation) to 1 (perfect correlation). Because it is a magnitude-based measurement, MS coherence is insensitive to phase shifts between signals.

We calculated MS coherence and the products of the cross-power spectra—phase difference $\phi_{xy}(\omega)$ and time delay $\tau_{xy}(\omega)$ —for receiver pairs grouped in source–receiver offset bins of 100 m, using a 25 m moving window from 0 to 1000 m and averaging within each bin. The outputs are average property distributions across both frequency and offset, allowing for comparison of the phase and amplitude of MEMS with 3CV and 3CH1 with DAS. Our analysis focused on the 5–100 Hz frequency band, which includes both surface and body waves.

3.4 Surface-wave analysis

We compared the recorded data using surface-wave analysis, a method based on the phase shift of dispersed surface waves (C.B. Park *et al.* 1999), which is a standard tool in near-surface exploration. This method, capitalizing on data redundancy arising from the large number of traces, enables a robust analysis of phase dispersion and, consequently, an estimation of subsurface shear-wave velocities (L.V. Socco *et al.* 2010; M. Wathelet *et al.* 2020). We used the method as a proxy for phase coherence and to test the hypothesis that an untrenched S-DAS survey is suitable for surface-wave analysis.

The data were sorted into receiver gathers, converted to ground velocity, but not correlated with the vibroseis sweep. The resorting was necessary to obtain coherent waveform displays, with no channel-dependent coupling distortions. The comparison of shot and receiver gathers with their corresponding f – k spectra is presented in Figs 6(a) and (b), showing the benefit of using common receiver gathers.

For each receiver gather, we applied offset windows of 300 m, sliding in 50 m increments from 100 to 1000 m. After computing the velocity-frequency spectra, we stacked spectra from repeated shots at the same locations and manually picked the dispersion curves at maximum spectral values. The picking was guided by a synthetic curve based on the velocity model in Table 1, ensuring selection of the fundamental mode. The array windows were consistent across all data sets, allowing direct comparison of the phase-shift response.

The retrieved dispersion curves for each array midpoint were inverted for shear-wave velocity using the nearest-neighbour inversion algorithm developed by M. Wathelet (2008) and implemented in the Geopsy package. The inversion was parametrized with three layers: shallow glacial till, weathered bedrock and bedrock, with a power-law velocity gradient in the second layer. The first layer was constrained to a maximum thickness of 20 m. The resulting models were evaluated based on the misfit between the input dispersion curve and the forward model response. We used the velocity model with the lowest misfit for further analysis. All models satisfied a misfit criterion of less than 5 per cent.

After applying the procedure to all extracted dispersion curves, we interpolated and smoothed the results using a 2-D Gaussian filter with a width of 301 m and a depth of 15 m. This resulted in smoothed 2-D shear-wave velocity profiles for the different data sets. For further considerations on the choice of offset window length, wavenumber resolution, inversion parametrization and investigation depth, we refer the reader to the work of Z. Wilczynski *et al.* (2024), who conducted a surface-wave study at the same site using the MEMS data set (active and passive).

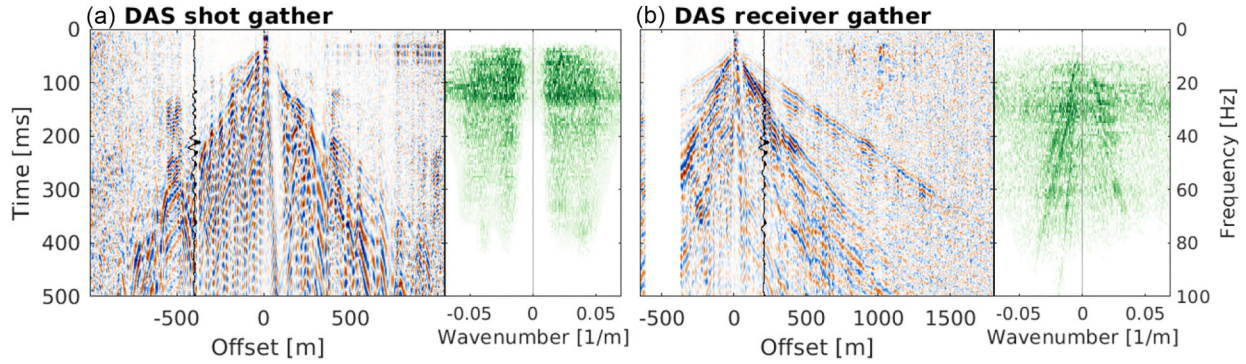


Figure 6. DAS data cross-correlated with a theoretical sweep (2–200 Hz, 18 s duration), normalized and sorted to: (a) common shot gather; (b) common receiver gather. For each panel, the plot on the right shows the corresponding f - k spectrum of the gather. All colour scales are normalized. The black wiggly traces overlaid on the gathers correspond to the shot or receiver sorting used in the other panel.

3.5 Ambient noise interferometry

Following previous studies on ambient noise interferometry in the area (Z. Wilczynski *et al.* 2024), we attempted to retrieve surface waves from ambient noise recordings. As shown there, noise correlations in nighttime data enabled the retrieval of Rayleigh waves propagating inline, originating from the south-eastern end of the profile, likely associated with a railway and a country road.

We used 8 hr of nighttime recordings for both the MEMS and DAS data sets, spanning the same time intervals from 22:00 to 06:00. Daytime passive data were not suitable for the ANSI experiment due to the presence of an operating active source. Due to an uneven spatial distribution of geophones and field performance issues, we did not attempt to calculate cross-correlation functions for the 3C geophone data sets. DAS data were decimated to 125 Hz, detrended and converted from differential phase rate to strain rate. Then, we used a sliding-window median filter with a window length of 150 m to remove common-mode noise and other time-invariant events. MEMS data were decimated to 125 Hz and detrended.

Then, for both data sets, we followed consistent processing steps outlined by H. Yao *et al.* (2006); we preprocessed the 1-hr-long traces using spectral whitening, divided the traces into frequency bands between 2 and 30 Hz, performed 1-bit normalization and finally stacked the bands back into broad-band, 1-bit normalized traces. The signals were then cross-correlated and stacked over all realizations to form cross-correlation functions (CCFs). We employed spectral whitening and 1-bit normalization to suppress amplitude variations and emphasize phase coherence between signals. The CCFs were sorted into virtual-source gathers, with one receiver serving as the source.

4 RESULTS

All data sets (MEMS, 3CV, 3CH1, DAS) were converted to particle velocity and are shown in Figs 5(a)–(d). The vertical-component records from the MEMS and 3CV sensors (Figs 5a and b) show good agreement, confirming the validity of the conversion. In contrast, the DAS gather (Fig. 5d) exhibits reduced data quality, particularly at very near offsets close to the source and at far offsets beyond approximately 1000 m. Notably, in the

horizontal geophone gather (Fig. 5c), the first breaks display opposite polarity on either side of the source, whereas in the DAS data, this polarity reversal is absent, which is an expected behaviour of DAS (E.R. Martin *et al.* 2021). The Rayleigh wave polarity, which is also expected to be reversed (E.R. Martin *et al.* 2021; Z. Wilczynski 2025), is more challenging to evaluate, as the waveform alignments changed in the surface wave time window (Fig. 5e, 200 to 250 ms).

4.1 Direct amplitude comparison

After converting all data sets to velocity, we compared the recorded amplitudes across the different sensors. The amplitudes were preserved during processing. Figs 7(a) and (b) show the normalized amplitude spectra averaged over all eight stations for 3CH1 and DAS for traces within two offset bins. For near offsets (25 to 100 m), with amplitudes scaled by approximately 0.65 (median value). DAS recorded higher amplitudes only in the lower frequency band below approximately 15 Hz. For the long offsets (100 to 1000 m), the scaling is approximately 0.35, whereas the spectrum's shape remains consistent. The comparison for vertical component sensors (MEMS and 3CV) is not presented, as the differences were negligible.

Figs 8(a) and (b) illustrates the RMS amplitude ratio between DAS and 3CH1, split across frequency bands representing different wavefield components: predominantly surface waves (5–35 Hz; Fig. 8a) and body waves and high-frequency energy (55–100 Hz; Fig. 8b). In the surface-wave band, although the results are dispersed, observations from individual stations follow horizontal trends. In general, DAS recorded lower amplitudes, with a ratio of mode 0.3 and a median around 0.5.

This relationship changes in the higher frequency band (Fig. 8b), where most amplitude ratios fall below 1. For velocity amplitudes below $1 \mu\text{m s}^{-1}$, however, a log-linear increase in the ratio emerges, indicating relatively stronger DAS amplitudes when geophone amplitudes are small. A similar trend appears in the post-sweep noise data. In Figs 9(a) and (b), the log-linear increase is evident across both frequency bands. Because the geophone-recorded noise generally remains below $1 \mu\text{m s}^{-1}$, the trend observed in Fig. 8(b) becomes even more pronounced, particularly in the higher-frequency band (Fig. 9b).

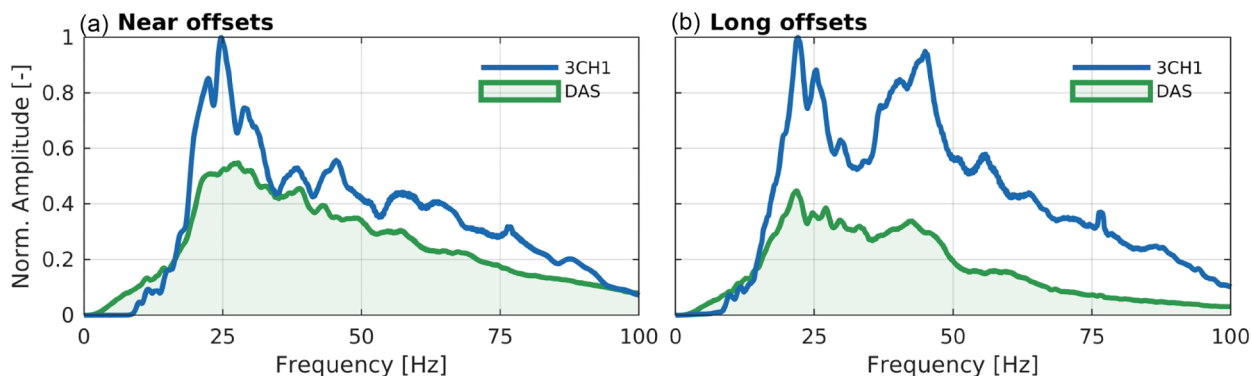


Figure 7. Normalized amplitude spectra of active source recordings of 3CH1 and DAS, averaged for all traces in the offset range of: (a) near offsets (25 to 100 m) and (b) long offsets (100 to 1000 m).

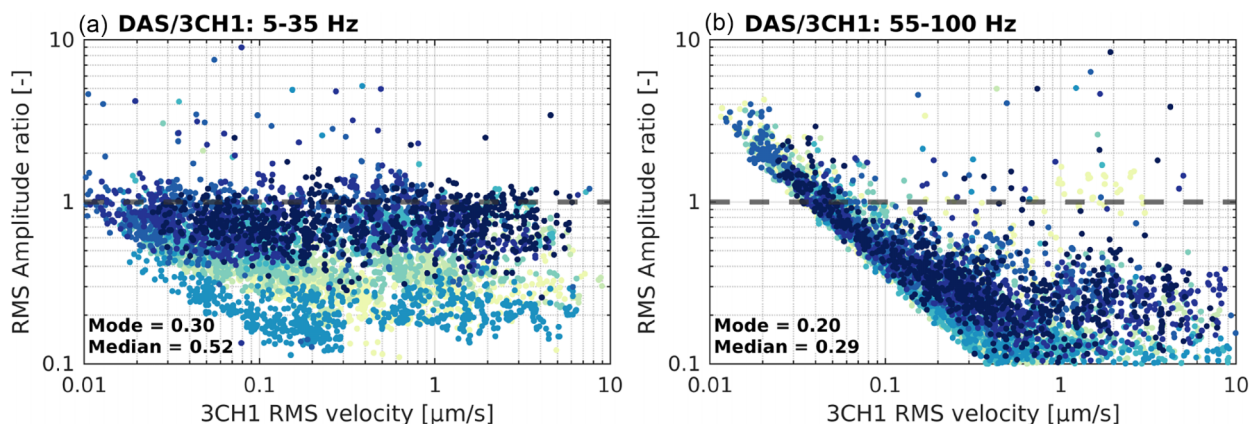


Figure 8. Ratio of ground velocity amplitudes recorded by DAS to 3CH1 during the vibroseis sweep (active source) in frequency bands: (a) 5 to 35 Hz (surface waves); (b) 55 to 100 Hz (body waves).

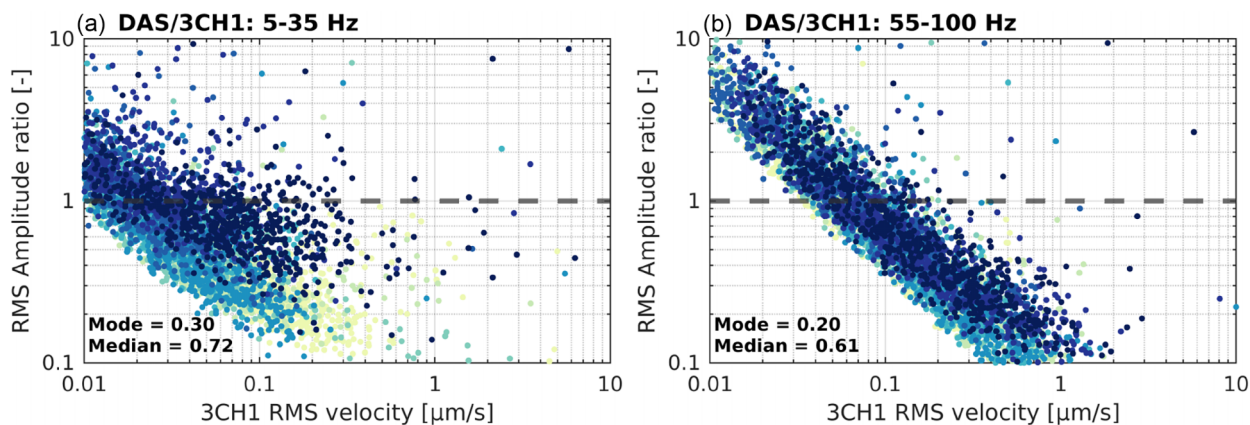


Figure 9. Ratio of ground velocity amplitudes recorded by DAS to 3CH1 during the post-sweep noise (inactive source), in the frequency bands: (a) 5–35 Hz (surface waves); (b) 55–100 Hz (body waves).

The comparison between the vertical component data sets (MEMS and 3CV) showed high agreement, with MEMS amplitudes generally slightly higher (median 1.05). The comparison between the horizontal geophone data sets (3CH1, 3CH2) yielded ratios close to unity, with the actual value varying by sta-

tion pair but remaining consistent across amplitudes. The DAS recordings were also compared to MEMS to cross-check the validity of the DAS-3CH1 comparison. It showed a similar anomalous behaviour in the low amplitude range of ground velocity, with the ratio increasing log-linearly.

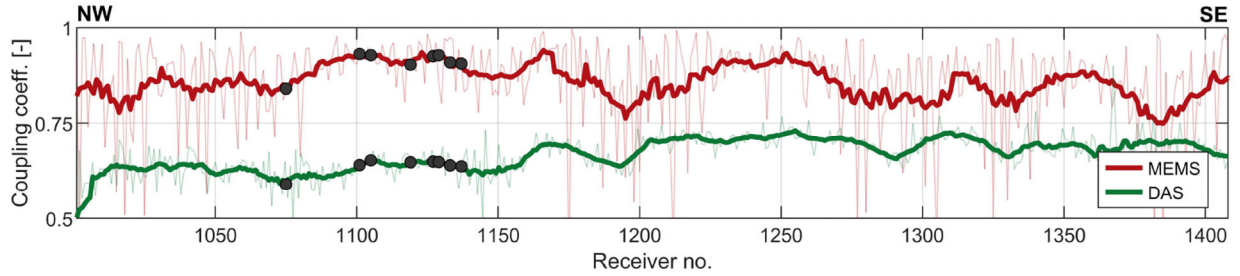


Figure 10. Sensor coupling coefficient for the MEMS and DAS data sets. Thin lines show channel-to-channel similarity; thick lines show an 11-point moving average. Black markers indicate channels selected for amplitude analysis (also shown in Fig. 3).

4.2 Coupling coefficient

The coupling coefficients for the DAS and MEMS data sets are shown in Fig. 10. The thin curves represent the channel-to-channel coefficients, while the thick curves show an 11-point moving average that highlights broader scale trends in trace similarity along the arrays.

For the MEMS array, coupling varies from channel to channel, but most coefficients remain above 0.75, indicating a generally high level of similarity between neighbouring sensors. In contrast, the DAS coupling curve is less variable, possibly attributed to the spatial averaging over the gauge length. However, DAS coefficients are predominantly below 0.75, implying lower channel-to-channel similarity and hence stronger spatial variability in effective coupling along the fibre.

4.3 Coherence and phase analysis

MS coherence, cross-power spectrum angle, and time delay are presented in Figs 11(a)–(f). Vertical-component data (MEMS and 3CV; Figs 11a, c, e) serve as a reference to evaluate coherence behaviour and phase stability with offset. For offsets up to 200 m, coherence is high, with values above 0.8 across the full 5–100 Hz band. Coherence decreases with increasing offset, particularly at higher frequencies, but remains above 0.6 at the largest offsets within the 20–60 Hz band. Phase differences range mainly between -30° and 30° , showing a slight delay in the surface-wave band (20–40 Hz) and a phase advance in the body-wave band (above 60 Hz). When expressed as time delay, the maximum reaches approximately 4 ms at far offsets and low frequencies, associated with surface waves. At shorter and intermediate offsets, the time delay generally remains below 2 ms, which is within one temporal sample of the recorded data.

The horizontal-component comparison between DAS and 3CH1 shows lower coherence (Figs 11b, d, f). Maximum coherence values remain below 0.8 and drop to 0.2 or lower above 40 Hz, even at offsets near 400 m. At very short offsets (0–100 m), coherence is lower than at slightly larger offsets, and coherence continues to decay with increasing offset, reaching values just above 0.2 at the farthest distances around 20 Hz. Although the overall decay pattern resembles that of the vertical-component pairs, the coherence values are substantially lower for DAS-3CH1 than for MEMS-3CV.

The estimated phase delay is strongly positive, indicating that DAS recordings arrive later than geophone recordings. The delay is most pronounced at higher frequencies and offsets of roughly 200–300 m, where phase differences approach 540° ($3\pi/2$). Phase differences decrease with both offset and frequency toward regions of lower MS coherence. After converting to time

delay (Fig. 11f), the surface-wave band shows delays of up to 40 ms. Overall, the DAS data exhibit systematic time variation that depends on both offset and frequency, consistent with the trends observed in the MS coherence values.

4.4 Surface-wave analysis

We present the results of surface-wave analysis using phase-shift MASW for 3CH1 and DAS. The results are summarized in Figs 12(a)–(f) as a panel of images.

The first row shows examples of frequency–velocity spectra used to extract the dispersion curves. The dotted line indicates the synthetic dispersion curve derived from the velocity model in Table 1, assuming a 4 m thick soil layer. The thin, sub-vertical white lines mark the array wavenumber resolution limits. Dispersion curves were extracted by following the spectral maxima. For 3CV data (a), the spectra exhibit a well-defined fundamental mode and are generally stable across frequency bins. In contrast, the DAS data (b) show degraded spectral quality, preventing reliable picking of dispersion curves at both low and high frequencies due to increased spectral noise. For several windows, we were unable to pick a dispersion curve accurately.

Panel (c) shows all picked dispersion curves for 3CH1 (background, light blue) and DAS (front, green). The dispersion curves span different frequency ranges, with the DAS curves not extending below 10 Hz and lacking resolution at the high-frequency end. In addition, the curves appear noisier than those for the 3CH1 data set.

Panel (d) allows for a statistical comparison between the extracted dispersion curves for the same receiver–offset configurations. The vertical axis shows the difference in velocity between picked points in a given frequency bin for the same station and the source–receiver offset. The solid line represents the mean, and the green-shaded area represents one standard deviation from the mean for all observations in the given frequency bin. The average velocity mismatch reached 200 m s^{-1} at the lower frequencies. It remained centred around 0 for observations at frequencies between 17 and 27 Hz, with an average standard deviation of 80 m s^{-1} . Thus, for velocities around 2000 m s^{-1} , the observed velocity difference remained in the 10 per cent misfit bounds.

The last row shows interpolated 2-D shear-wave velocity sections from the inversion of the dispersion curves. The sections differ especially at the beginning of the profile, where fewer curves are extracted. However, in the central part of the profile (800–1100 m), where data density is highest, the DAS results (e) broadly agree with 3CH1 (f). In both data sets, several consistent subsurface features can be identified: a low-velocity

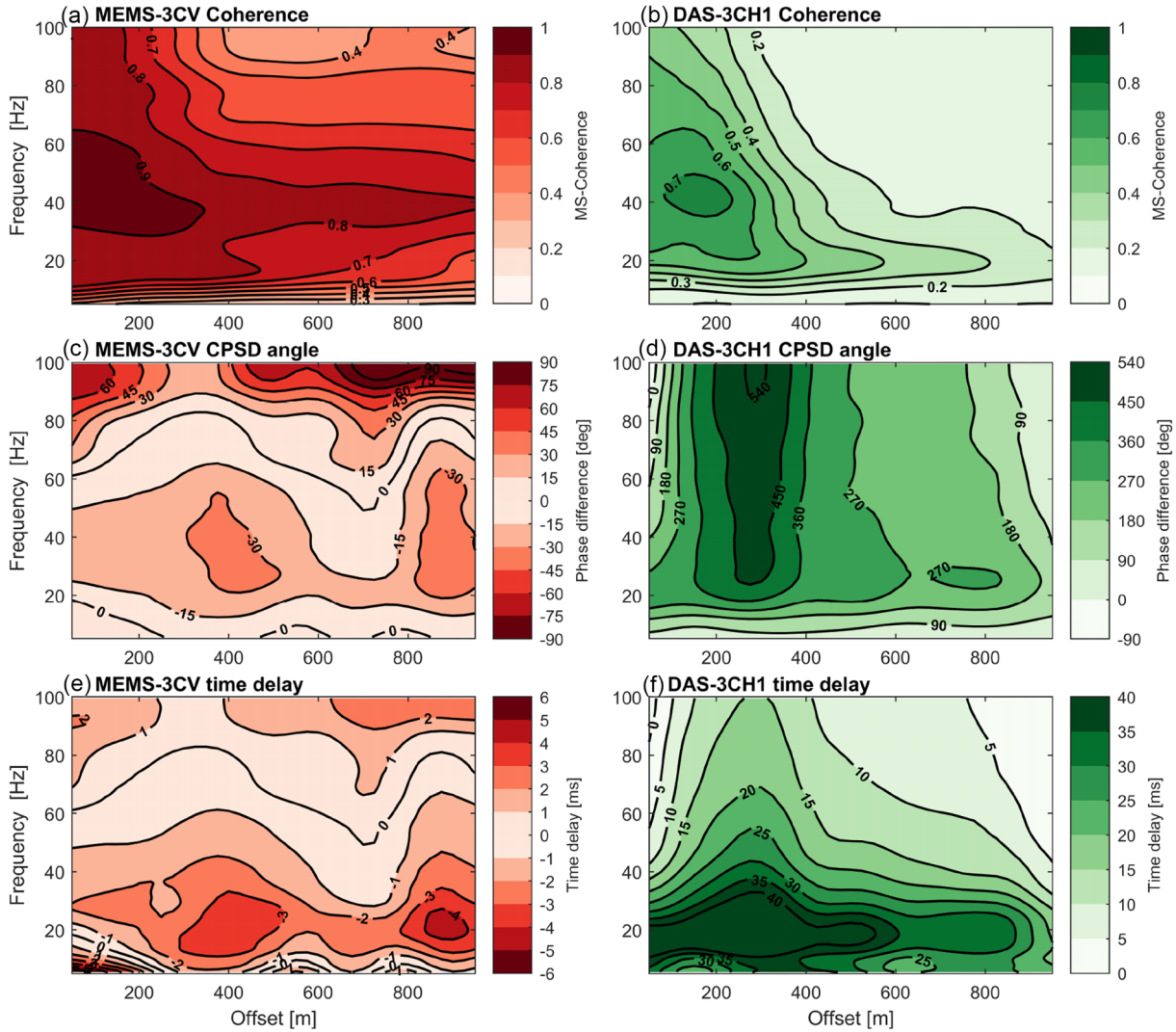


Figure 11. Results of MS Coherence and cross-power spectrum analysis obtained using a 100 m window, sliding with an increment of 25 m for source–receiver offsets between 0 and 1000 m. Results are averaged within each offset bin. The left column shows comparisons of MEMS with 3CV (a, c, e), and the right column shows comparisons of DAS with 3CH1 (b, d, f). The rows present the estimated quantities: magnitude-squared coherence (a, b), phase difference (c, d) and time delay (e, f).

anomaly centred around 900 m, reaching depths of approximately 75 m; a high-velocity zone at the end of the profile (1100–1500 m), extending to shallow depths; a high-velocity region at the beginning of the profile (around 750 m).

4.5 Ambient noise interferometry

Cross-correlation of the nighttime ambient noise recorded by the MEMS and DAS systems produced a large number of cross-correlation functions. For the MEMS data, these results were previously organized into virtual shot gathers and used for surface-wave analysis by Z. Wilczynski *et al.* (2024). In Fig. 13a,b, we compare the cross-correlation functions (CCFs) of only one station (no. 7) with those of all other stations for both the MEMS and DAS data sets.

For MEMS, we successfully retrieved a Rayleigh wave with an apparent velocity of approximately 2400 m s^{-1} , arriving from the southern end of the profile. The arrival is strong and coherent across offsets, with minimal noise interference. Most of

the energy is concentrated between 5 and 15 Hz. In contrast, for DAS, cross-correlation failed to retrieve a clear medium response. The most prominent feature is a near-offset horizontal (time-invariant, infinite velocity) unphysical event. Beyond offsets of roughly 250 m, no coherent arrivals were observed. The DAS energy is distributed within a broader frequency band of 5 to 25 Hz, with three distinct peaks.

5 DISCUSSION

In the following subsections, we discuss the comparative results for amplitude and phase, assess the fidelity of our findings, and offer practical considerations for the field deployment of untrenched surface-DAS.

5.1 Coupling effects

The coupling coefficient of DAS (Fig. 10) shows channel-to-channel similarity and is, in general, significantly lower than

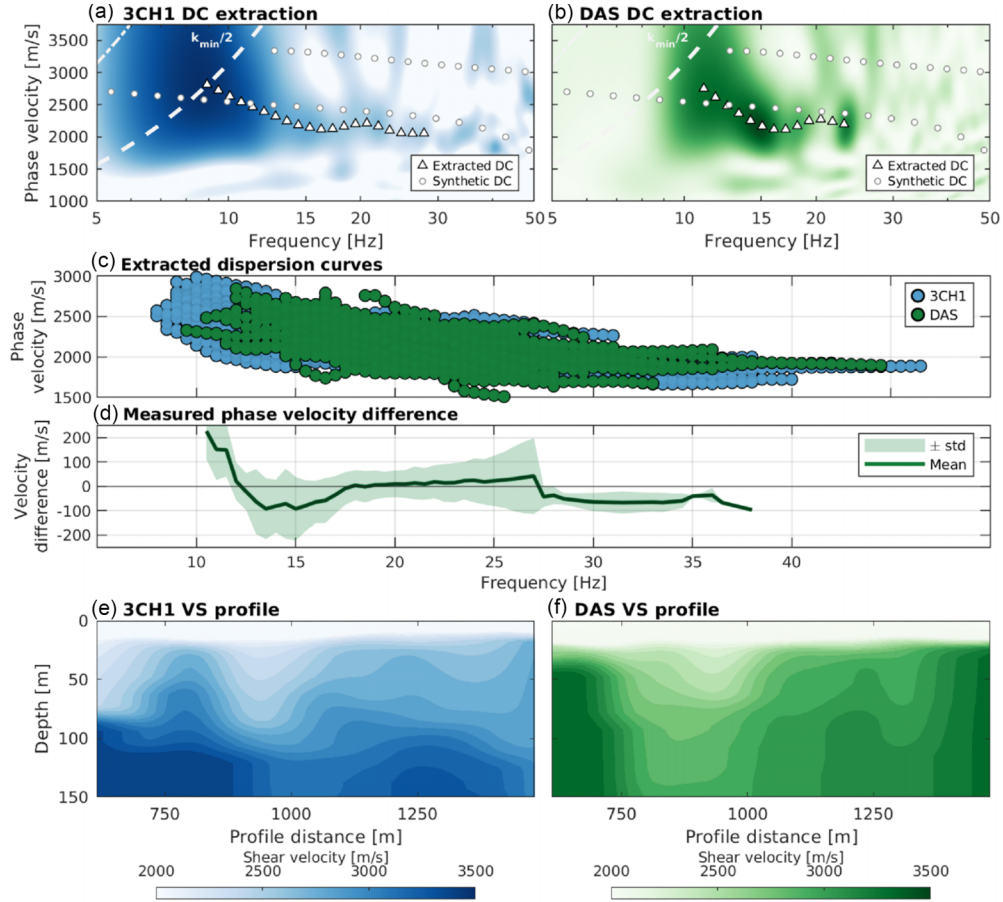


Figure 12. Summary of the phase-shift MASW surface-wave analysis for the recording systems: 3CH1 (left, blue) and DAS (right, green). (a) and (b) Example frequency–velocity spectra with overlaid synthetic dispersion curve (white dotted line) based on the *a priori* velocity model (Table 1), array wavenumber resolution limits (dashed white lines), and picked dispersion curves (triangles). (c) Extracted dispersion curves for both data sets. (d) Measured velocity difference for the same receiver–offset configurations. The shaded area indicates one standard deviation from the mean. (e) and (f) Interpolated 2-D shear-wave velocity profiles obtained from the dispersion curves inversion.

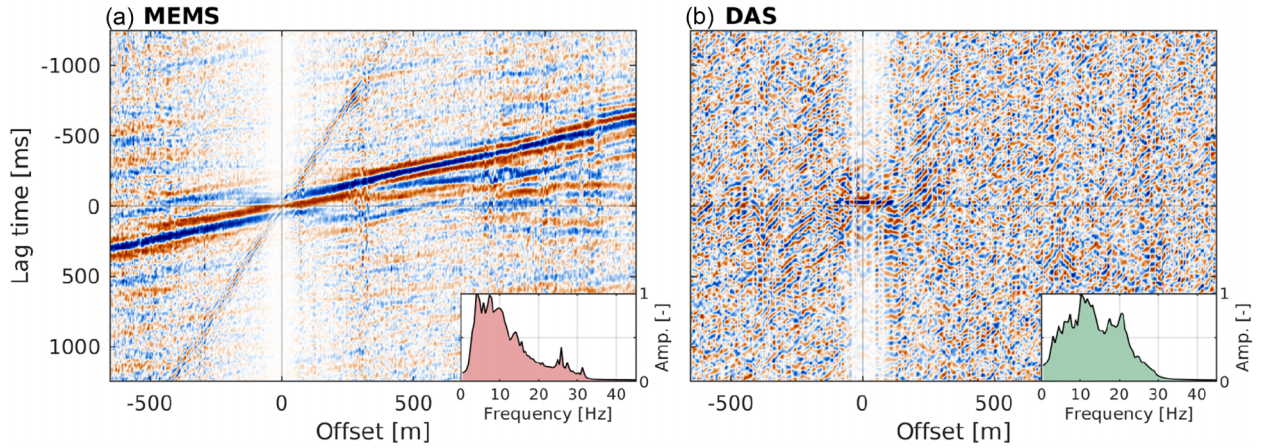


Figure 13. Comparison of cross-correlation functions with a single station for 8 hr of nighttime recording for: (a) MEMS data set, showing a clear Rayleigh-wave arrival; (b) DAS data set, contaminated with a strong, space-invariant noise. The corresponding averaged normalized amplitude spectra are displayed in the corner.

that of MEMS, providing useful context for interpreting DAS performance in amplitude analysis and phase-based methods.

The variability in coupling quality is likely attributable to the deployment method used in this study, that is, an untrenched

fibre covered with sand. This interpretation is consistent with prior theoretical discussions of DAS coupling and strain-transfer effects (T. Reinsch *et al.* 2017; N.L. Celli *et al.* 2023), as well as empirical observations that improved coupling can be achieved by

applying additional loading (A. Pandey *et al.* 2023; B. Brodic *et al.* 2023), or by mechanically bonding the fibre to the ground, for example, via freezing (J. Yang *et al.* 2022; T. Hudson *et al.* 2025). These strategies mitigate coupling heterogeneity by increasing normal stress at the fibre-soil interface to enhance strain transfer, introducing bonding or confinement to reduce fibre motion relative to the surrounding medium. In our study, the untrenched deployment with only a thin sand cover likely provided insufficient normal load and negligible bonding, explaining the observed channel-to-channel coupling variability and relatively low coupling quality.

5.2 Amplitude behaviour

The amplitude analysis revealed consistent differences between DAS and conventional sensors, primarily due to differences in coupling conditions and instrument responses. While in the surface-wave band, the amplitudes were consistently scaled down; in the high-frequency band, signals showed systematic deviations.

For lower geophone amplitudes, the DAS-3CH1 amplitude ratio exhibited a log-linear increase, especially at high frequencies (Figs 8b and 9b). The trend is more likely a result of high-frequency noise from loose coupling (N.J. Lindsey *et al.* 2020), common-mode noise, residual bias remaining after weighted moving-average filtering, and system-specific high-frequency noise. Due to poor fibre coupling, the amplitudes can be selectively amplified (N.L. Celli *et al.* 2023), or attenuated through poor strain transfer (T. Reinsch *et al.* 2017). Additionally, interface waves can further distort the waveforms and, consequently, the trace RMS amplitudes.

Notably, amplitude ratios between DAS and 3CH1 exhibit approximately linear behaviour in the surface-wave frequency band (Fig. 8a), and for geophone velocities above $1 \mu\text{m s}^{-1}$ in higher frequency bands (Fig. 8b). However, the observed amplitude ratios are strongly station-dependent. While some stations show good agreement in low-frequency bands, most yield ratios below unity. This, however, is to be expected, as DAS amplitudes are filtered in wavenumber by both the gauge length and the moving Hann window, the latter scaling the amplitudes by approximately 0.85. Given the median amplitude ratio of 0.52 (Fig. 8a), we argue that an accurate amplitude recovery is achievable from high-quality DAS data, but it depends strongly on the fibre coupling.

MS coherence plot (Fig. 11b) shows that the DAS-recorded waveforms are incoherent with the horizontal geophone, particularly at higher frequencies and long offsets. This further confirms the low sensitivity of untrenched DAS to low-amplitude vibrations. Nevertheless, the low frequencies remain the least distorted part of the waveforms, which potentially explains the successful surface-wave experiment. This aligns with the theoretical framework of T. Reinsch *et al.* (2017), who showed that the strain transfer favours lower frequencies and longer wavelengths.

Comparison of the amplitude spectra (Figs 7a and b) shows that for both near (25 to 100 m) and long (100 to 1000 m) offsets, the overall shape of the amplitude spectra is similar. However, DAS amplitudes are consistently scaled down, with the scaling being stronger with increased offsets. These observations suggest that the difference in recorded amplitudes depends on the source distance, with DAS showing reduced sensitivity at

larger offsets, where wavefront amplitudes are reduced by spherical divergence and scattering. This is additionally supported by magnitude-squared coherence decaying with increasing offsets (Fig. 11b). While it is natural for the coherence to drop with distance (Fig. 11a), the low coherence values and a quicker decay indicate that DAS does not coherently record the low amplitude arrivals.

Higher amplitudes were observed only for the lowest frequencies, below approximately 15 Hz. The elevated low-frequency amplitudes in DAS data can potentially be attributed to selective amplification observed numerically by N.L. Celli *et al.* (2023), or to the monochromatic, space-invariant noise (Fig. 6a), that becomes less apparent when resorting to common receiver gathers.

The two components of the amplitude analysis—amplitude spectra and trace-to-trace RMS amplitude comparisons—yield seemingly contradictory results for the DAS data. The averaged amplitude spectra (Figs 7a and b) show a consistent amplitude scaling across frequencies. In contrast, the RMS amplitude ratio plots (Figs 8a and b) indicate stronger (but station-dependent) agreement between DAS and geophone amplitudes in the low-frequency band (5–35 Hz), while the high-frequency band is affected by a log-linear trend in the amplitude ratios.

This apparent discrepancy arises from the differing sensitivities of the two methods. While the amplitude spectra average out spatial variability and local anomalies, the RMS amplitude ratios are computed trace by trace in the time domain, making them more sensitive to local spatial variations (e.g. coupling or noise) but lacking frequency resolution. However, the log-linear trend appears predominantly for low-energy traces (RMS amplitude below $0.5 \mu\text{m s}^{-1}$). This effect can be linked to broadband noise contamination in the DAS data, visible in the shot gather display in Fig. 6(a). We identify it as DAS-specific noise linked to the acquisition setup. As its amplitude is high across all frequencies, it becomes more significant for low-amplitude signals (body waves) than for high-amplitude events (surface waves).

The primary findings from amplitude analysis are: (1) DAS recordings exhibit substantially elevated high-frequency noise yet maintain reasonable agreement with reference sensors in the surface-wave band; (2) the log-linear amplification of low-energy traces ($< 1 \mu\text{m s}^{-1}$), particularly for the high frequency band, shows the system-specific noise, possibly attributed to the poor coupling; (3) strong station-to-station variability, particularly evident in 3C geophone and DAS data; (4) DAS amplitudes are systematically attenuated through combined effects of incidence angle directivity, gauge-length sinc filtering, and Hann-window weighting; (5) DAS loses the amplitude fidelity and coherence in higher offset range.

5.3 Phase fidelity and phase-based near-surface methods

The phase difference and the corresponding time delay provide insight into the phase fidelity of the signals. For the MEMS and 3C sensors, the observed phase differences generally fall within a range equivalent to 2 ms, matching the temporal sampling of the recorded data (3CV, 3CH1). This can be partly attributed to a static shift caused by small variations in source distance and by the presence of a low-velocity near-surface layer. In contrast, the delay between the DAS and 3CH1 signals is more pronounced within the surface-wave band.

In general, low coherence indicates that the cross-power spectra are unreliable. Notably, the largest delays occur when MS coherence values are high. This suggests that reliable coherence is mainly obtained at low frequencies and short offsets, and that the DAS data exhibits a systematic time delay of approximately 20–40 ms. Part of this delay may stem from acquisition-related effects: specifically, the imprint of receiver spacing and the influence of spatial integration on noisy data. However, it is also consistent with the findings of N.L. Celli *et al.* (2023), who demonstrated that certain surface-wave modes appear delayed by a thin sediment cover and imperfect strain transfer. We presume that the observed phase delay is the combined result of both factors.

However, the phase fidelity between DAS and reference sensors was sufficient to retrieve meaningful dispersion curves (after resorting to common receiver gathers). The frequency range was limited compared to 3CH1, but the DAS-extracted phase velocities differed from 3CH1 for the same receiver-offset configurations by less than 10 per cent, with the mean mainly centred around 0. The lower extracted phase velocity in the frequency range between 12 and 17 Hz likely results from the systematic phase delay observed in the time delay plot (Fig. 11f).

The surface-wave analysis results indicate that untrenched DAS may remain viable for surface-wave analysis under suboptimal coupling conditions, as previously suggested by N. Harmon *et al.* (2022). Even with reduced resolution, the obtained shear-velocity sections highlight the same low-velocity anomaly, providing an essential insight into the geological setting at this site. Similar results for conventional sensors were shown to be valuable in the mineral exploration context in this area (M. Papadopoulou *et al.* 2019; Z. Wilczynski *et al.* 2024). In surface-wave analysis, as the method uses the relative shifts between particular phases, the system's phase change (Figs 1a and d) does not affect the final result.

A key practical finding is that resorting the data into common-receiver gathers (Figs 6a and b) substantially improves waveform coherence by selecting traces from the same channel, thereby ensuring more consistent coupling. This step proved essential for the successful application of MASW using untrenched DAS data. The benefit of resorting is clear in the f – k spectra (Figs 6a and b), where coherent waveforms arise in the receiver gather.

Our attempt to retrieve surface waves from noise cross-correlations was unsuccessful, despite applying 1-bit normalization and spectral whitening to suppress amplitude effects and using a median filter to remove common-mode noise. The resulting cross-correlation functions remained strongly contaminated by time-invariant events, similar to the results of (V.R. Tribaldos *et al.* 2021, fig. 15.5a). We chose to cross-correlate the strain-rate data directly, avoiding unnecessary integration and filtering. However, integrating the strain rate to strain and applying a Hann window could have further reduced the impact of common-mode noise by attenuating low-wavenumber energy (Figs 1f and 5).

We attribute DAS's reduced performance in ambient noise processing mainly to high channel-to-channel coupling variability, leading to an incoherent response across channels easily observed in a shot gather display (Fig. 6a). When cross-correlating channels with waveform recordings distorted by poor coupling, the cross-correlation function is expected to appear noisy, inhibiting the retrieval of the medium's response.

Additionally, as shown in Figs 8 and 9, at low ground velocity amplitudes (below $1 \mu\text{m s}^{-1}$ in the geophone data), the DAS recordings showed significantly higher amplitudes, which we interpret as system-specific noise. Since ambient noise typically produces only low ground vibrations (see Fig. 9), the DAS traces are likely contaminated by incoherent noise, possibly related to the coupling layer material (N.L. Celli *et al.* 2023). Such contamination prevents the reliable reconstruction of the Green's functions and the retrieval of the true wave arrivals.

A potential improvement would be to utilize daytime noise recordings, which yield higher ambient vibration amplitudes (X. Zeng *et al.* 2017) and could exceed the system-specific noise, allowing successful cross-correlation. Unfortunately, we did not collect any daytime data before or after the active-source shooting, and therefore, we could not process daytime ambient vibrations.

Future work on ambient noise recorded by DAS could explore suppressing horizontal coherent noise through more targeted median filtering (e.g. S.K. Bakku 2015; V.R. Tribaldos *et al.* 2021; L. Gyger *et al.* 2024) or frequency–wavenumber (f – k) filtering (M.R. Fernández-Ruiz *et al.* 2020). Further, the signal-to-noise ratio might be enhanced by applying phase-weighted stacking (M. Schimmel & H. Paulssen 1997; V.R. Tribaldos *et al.* 2021) to the cross-correlation functions, preserving only coherent energy. These refinements, however, are beyond the scope of this study. Our findings show that, when standard cross-correlation is applied directly to strain-rate data, the untrenched DAS does not yield satisfactory surface-wave retrieval at this site and requires more complex processing strategies.

5.4 Results fidelity

Throughout this study, we assumed that the conversion from acceleration to velocity was carried out using an ideal integrator. This assumption holds valid only within a specific frequency range, where a low-pass filter can approximate ideal integration (S. Thenozhi *et al.* 2012). To ensure stability, we carefully selected the integration parameters to mitigate low-frequency instabilities. Corner frequencies below 1.5 Hz were found to introduce nonlinear trends in the resulting velocity estimates. To ensure a stable process, we limited the analysis to above 5 Hz. The adopted parameters yielded consistent, stable velocity waveforms (Figs 5a and e).

It is important to note that the spatial integration method proposed by (A. Trabattoni *et al.* 2023) does not approximate a true integrator, unlike the temporal integration used to convert MEMS data, but instead performs a cumulative sum. This makes the approach more susceptible to noise biases, which can affect the cross-power spectra and reduce the accuracy of the phase estimates.

When comparing DAS amplitudes to other systems, we did not correct for amplitude attenuation, as this is not trivial and requires f – k rescaling (T. Daley *et al.* 2015; N.J. Lindsey *et al.* 2020), which introduces additional signal modifications (P. Paitz *et al.* 2020). On average, the Hann window proposed by A. Trabattoni *et al.* (2023) rescales the amplitudes by 0.85, with the exact value depending on the waveform's apparent velocity and the signal frequency. A similar correction applies to the gauge length sinc transfer function. Additionally, the moving average smears transient noise across channels, introducing errors in the RMS amplitude estimates for traces within the window.

Another amplitude factor is related to the directional sensitivity of DAS. As the strain record is projected along the fibre, the amplitudes undergo scaling with a $\cos^2(\theta)$ factor, where θ is the incidence angle of the incoming wave (A. Mateeva et al. 2014; E.R. Martin et al. 2021; Z. Wilczynski 2025). It gives the maximum normalised difference of 0.25 observed at 60 degrees. For a Rayleigh wave propagating inline ($\theta = 0$ degrees), the amplitudes are not decreased, which is also the case for a geophone, with its incidence sensitivity being $\cos(\theta)$. For P waves arriving at an angle θ to the surface, the recordings experience a higher amplitude decrease for DAS than for geophones. Moreover, for SV-waves, the DAS scaling depends on the $\sin(2\theta)$ factor (S.K. Bakku 2015; E.R. Martin et al. 2021; M.E. Willis 2022; Z. Wilczynski 2025). It introduces more incidence-dependent scaling when compared to the horizontal geophone, with a polarity flip between 90 and 270 degrees. The scaling of both longitudinal and shear waves may explain some of the discrepancies in the higher frequency band (Fig. 8a).

As shown in the averaged amplitude spectra (Fig. 7b), DAS records lower amplitude values, with amplitudes scaled relative to the horizontal radial geophone. This reduction can potentially be attributed to the previously discussed amplitude attenuation factors. For an incidence angle of 60 degrees and the application of the Hann window during conversion, the amplitudes would be scaled by about 0.75 and 0.85, resulting in an overall factor of approximately 0.63, which could partially explain the observed amplitude difference. However, this remains a simplified explanation of a more complex effect, as the surface waves do not arrive at a 60-degree inclination, and the Hann window's attenuation effect is wavenumber-dependent and nonlinear, particularly damping lower frequencies.

We used RMS amplitude as our main amplitude attribute, which effectively captures average signal energy and system noise, and is well-suited to continuous vibroseis signals. However, it does not reflect peak amplitudes, which are essential for earthquake seismology (I. Lior et al. 2023; A. Trabattori et al. 2023). We avoided peak amplitudes because of their high sensitivity to transient noise sources.

Our investigation assumed that the sensors were perfectly collocated, which is impossible in practice. While accelerometers and geophones were collocated within a few centimetres, the fibre was deployed roughly 1 m away, typically on a different surface and covered with an additional layer of unconsolidated material, which may partially explain the time delay (Figs 11e and f). Additionally, the location of the DAS channels is determined by tap tests, which provide the fibre-optical path length at a given point. With the tap test conducted at only several points, the channel-coordinate interpolation error increases, even in a nearly linear deployment. We have inspected the quality of the channel numbering by comparison to the accelerometer data after vibroseis sweep correlation (as in Fig. 5). We did not observe a significant shift in the zero-offset trace, allowing us to assume that the DAS channels' coordinates were accurate within ± 5 m.

5.5 Implications for practitioners

We evaluate the results in the context of the practical implications of using untrenched S-DAS. Differences in recorded particle velocity amplitudes between systems arise from several

amplitude-scaling factors, including coupling-dependent strain transfer (N.L. Celli et al. 2023; T. Hudson et al. 2025), gauge length effects (T. Dean et al. 2016a; M.E. Willis 2022), and incidence angle sensitivity (A. Mateeva et al. 2014). These effects may pose significant limitations for surveys that rely on preserving absolute amplitude information (e.g. AVO, FWI). Potential mitigation strategies include correcting for coupling effects as proposed by T. Hudson et al. (2025) and compensating for gauge length filtering via signal rescaling in the f - k domain.

In this experiment, we observed a strong log-linear amplitude trend in the body-wave frequency band (55–100 Hz), which enhanced the recorded amplitudes of small ground velocity signals (Figs 8b and c). This trend is likely associated with high levels of DAS-specific noise, impeding the extraction of the medium's response from ambient vibrations. Additionally, it can compromise body-wave arrivals and pose a significant limitation for reflection-imaging applications.

The untrenched deployment also led to an uneven strain transfer, resulting in variable coupling quality between stations (Figs 10 and 8a). Of the 420 available channels, only approximately 220 met the usable data standard, with the remaining channels exhibiting quality below the exploration standard. Uneven coupling is an inherent limitation of this deployment method and should be anticipated during survey planning and data quality assessment. However, resorting the data to a common receiver (Figs 6a and b) significantly improved the quality of the gathers and enabled a successful MASW application.

The strong coherence decay with increasing offsets (Fig. 11b) and the log-linear amplitude increase for low vibrations (Fig. 8b) suggest low DAS sensitivity to low-amplitude signals, limiting the applicability of this deployment to high-amplitude sources (active-source surface-wave studies at near offsets).

The used gauge length of 10 m allows for a reasonably flat response for velocities around 2500 m s^{-1} below 100 Hz (Fig. 1d), and for higher velocities at higher frequencies. In the hardrock conditions of high and medium velocities, this gauge length is suitable. However, when working at soft-sites or sedimentary basins, the velocities are generally lower, which will introduce a sinc zero-crossing at lower frequencies. This is an essential factor to consider when planning a survey.

Despite the anticipated improvement in long-wavelength strain transfer associated with a thin fibre coupling layer (T. Reinsch et al. 2017), cross-correlation of the ambient noise data did not recover low-frequency surface waves. This result indicates that two conditions must be satisfied for a successful interferometric experiment. First, the system must be sufficiently sensitive to low-amplitude vibrations to capture the ambient noise wavefield. Second, the fibre coupling must be spatially uniform to ensure a consistent response to the incoming wavefield. Both requirements can be achieved through weight loading or mechanical bonding, as demonstrated in previous studies (J. Yang et al. 2022; B. Brodic et al. 2023; A. Pandey et al. 2023; T. Hudson et al. 2025; V. Stender et al. 2026).

In experiments with limited control over the coupling fibre, for example, when using dark or telecommunication fibres, more advanced pre-processing and post-processing strategies may be employed to improve the results of ambient noise interferometry, such as those proposed by V.R. Tribaldos et al. (2021).

Despite the apparent limitations, we were able to use the DAS data for surface-wave imaging, and L. Gyger *et al.* (2025) successfully imaged the main iron-oxide mineralization as a set of strong reflections. Thus, even though the data quality is lower, it may still be sufficient for particular targets or purposes. Nevertheless, the most effective recommendation remains to ensure proper coupling, which can be achieved by increasing the thickness of the sand overburden (T. Reinsch *et al.* 2017), soil compaction (B. Brodic *et al.* 2023), freezing (J. Yang *et al.* 2022; T. Hudson *et al.* 2025), or pinning (N. Harmon *et al.* 2022).

With recent advances in nodal sensor technology enabling ultra-high-resolution recording (T. Pertuz & A. Malehmir 2023), the spatial sampling advantage of DAS is less pronounced. However, DAS retains a key benefit in scalability. Expanding nodal arrays increases costs per unit, whereas the number of DAS channels is mainly determined by the fibre length. This gives DAS greater flexibility to scale acquisition over long distances or to increase spatial sampling compared to nodal arrays.

6 CONCLUSIONS

We assessed the performance of surface-DAS by comparing its amplitude and phase fidelity with 3C-geophones and MEMS-based accelerometers. While S-DAS showed elevated noise at low amplitudes and reduced high-frequency sensitivity, it performed reliably in the surface-wave frequency band, where amplitudes scaled linearly with conventional sensors under good coupling conditions.

By resorting the data to common-receiver gathers, the phase-based surface-wave dispersion analysis produced consistent results across all sensor types, and the extracted phase velocities closely matched those obtained from conventional sensors. This demonstrates that DAS can provide valuable subsurface information even with imperfect coupling. However, the quality of the DAS data varied considerably along the fibre, with only about half of the channels suitable for interpretation. Furthermore, the untrenched S-DAS response to ambient noise interferometry was unsatisfactory due to variable receiver coupling and low strain transfer, suggesting that ambient noise studies would benefit from deployment strategies providing a better coupling.

Despite limitations, untrenched DAS offers a practical complement or alternative to traditional sensors for applications focused on surface waves, or where a large channel number is prioritized. Ultimately, the decision to use surface-DAS should be made on a case-by-case basis, weighing its limitations and benefits against the survey's specific goals, site conditions, and hardware availability.

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DATA AVAILABILITY

Research data will be available after the data set potential has been explored by the project collaborators. Currently, there is a 3-yr embargo on the data. For further instructions, contact alir.eza.malehmir@geo.uu.se.

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